Pressure-Robust Error Estimates for a Stabilized BDF2–IMEX Navier–Stokes Scheme

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Abstract

We analyse a fully discrete scheme for the incompressible Navier–Stokes equations built from three ingredients: Scott–Vogelius elements on Alfeld splits, which give exactly divergence-free velocities; a continuous interior penalty (CIP) stabilization of the convective derivative with three jump terms; and a linearly implicit BDF2 method. The convective term and the stabilization are both evaluated with the extrapolated velocity $2u_h^n - u_h^{n-1}$, so every time step costs a single linear solve. The scheme is unconditionally energy stable, and the stability bound neither depends on ν nor contains Gronwall factors. For smooth solutions we prove an L^2 velocity error of order $\left((\nu+h)h^{2k}+\Delta t^4\right)^{1/2}$, with constants that depend neither on the pressure nor on negative powers of ν. When $\nu \lesssim h$ this is $O(h^{k+1/2}+\Delta t^2)$, the best rate known for ν-independent L^2 estimates. The argument is complete in two dimensions for $k \in \{2,3\}$; in three dimensions it relies on a potential-recovery property that is still open. The normalized stabilization needs upper and lower bounds on the extrapolated velocity. These follow from a preliminary robust estimate if $\Delta t \le c_0 h^{1/2+\sigma}$, and a clamped normalization removes this condition from the velocity estimate. For the pressure we obtain a robust bound of order $h^{k-1}+\Delta t^2/h$ in the discrete $\ell^2(L^2)$ norm. Numerical experiments in two dimensions, and in three dimensions with a simplified variant of the scheme, show the predicted rates and the robustness with respect to the pressure and the viscosity.

Keywords: incompressible Navier–Stokes equations; pressure robustness; convection robustness; continuous interior penalty; Scott–Vogelius elements; BDF2; IMEX; error analysis.
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1 Introduction

New references [BH04, GLS19, Lin14, LM16, SV85]: check volume and page numbers.

Simulating incompressible flow at high Reynolds number remains hard for finite element methods, and two effects are largely responsible. The first comes from the way velocity and pressure are coupled. For classical inf-sup stable pairs such as Taylor–Hood, the velocity error bound contains the term $\nu^{-1}\inf_{q_h}\|p-q_h\|_0$. A large or complicated pressure can therefore spoil the velocity, even when the velocity itself is smooth [21, 34, 35]. Methods whose velocity error does not see the pressure are called *pressure robust*. The second effect is convection. Once ν is small compared with the mesh size, the Galerkin solution develops spurious oscillations, and the usual error bounds

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blow up through Gronwall factors such as $\exp(CT\nu^{-1})$ or through explicit powers of ν^{-1}. We call a method *convection robust* (or Reynolds robust) if its error constants contain no inverse powers of ν. Why both properties matter in practice is discussed in [33, 17].

Pressure robustness comes for free with exactly divergence-free elements. On Alfeld (barycentric) splits, the Scott–Vogelius pair [36, 2, 23, 18] produces discrete velocities that are divergence-free pointwise. If the error equation is tested with such a function, both the continuous and the discrete pressure drop out. Already the plain Galerkin method then gives L^2 velocity bounds of order h^k that are robust with respect to both the pressure and ν [22, 17]. To do better when convection dominates, some stabilization is needed; the best rate known for ν-independent bounds is $h^{k+1/2}$ [17]. It was first proved for $H(\mathrm{div})$-conforming discontinuous Galerkin methods with upwinding [19, 4, 20]. For H^1-conforming divergence-free elements, it has been obtained with a residual stabilization of the vorticity equation for the Oseen problem [1] and with an SUPG-curl stabilization for the time-dependent Navier–Stokes equations [5].

Continuous interior penalty (CIP) methods [32, 9, 8] stabilize by penalizing jumps of the convective derivative across interior faces. They are symmetric, they leave the velocity–pressure coupling untouched, and they need no residual of the strong equation. With divergence-free elements, however, the classical gradient-jump penalty is not enough for the $h^{k+1/2}$ rate. The reason lies in the consistency argument, which has to recover volume terms from face jumps in a space of potentials. Barrenechea, Burman, Cáceres and Guzmán [3] overcame this for the steady Oseen problem. Their three-term CIP form penalizes the jumps of $(\beta\cdot\nabla)u_h$, of $\nabla\times((\beta\cdot\nabla)u_h)$, and of the gradient of the latter. Combined with the C^1 Clough–Tocher potential space, it yields pressure- and convection-robust bounds of order $h^{k+1/2}$ in two dimensions. García-Archilla and Novo [16] then carried the method over to the semi-discrete Navier–Stokes equations, using the discrete velocity as transport field and normalizing the stabilization by $\|u_h\|_{0,\infty}^{-1}$.

What is still missing is a time discretization whose analysis does not bring back a dependence on ν^{-1} or on the pressure. A fully implicit scheme would require a nonlinear solve in every step, which is especially expensive here because the stabilization depends nonlinearly on the solution, both through its transport field and through its normalization. A linearly implicit BDF2 scheme avoids this. We replace the transport field by the second-order extrapolation $\tilde{u}_h^{n+1} = 2u_h^n - u_h^{n-1}$, so that each step amounts to one linear system with a time-dependent matrix. Robust analyses of IMEX-BDF2 and BDF-q schemes are available for grad-div stabilized methods [15, 14]; they give ν-robust bounds of order h^k. Space–time DG methods with $H(\mathrm{div})$-conforming velocities and upwinding were analysed in [6], and a Reynolds semi-robust hybrid method in [7]. Implicit–explicit multistep treatments of the CIP term itself are studied in [10].

Contribution. In this paper we analyse the three-term CIP method of [3, 16] with Scott–Vogelius elements and a linearly implicit BDF2–IMEX time discretization, in which both the convective term and the stabilization are evaluated with the extrapolated transport field $2u_h^n - u_h^{n-1}$. Our main results are as follows.

(i) The scheme is unconditionally energy stable; the bound holds for every Δt, does not depend on ν, and contains no Gronwall factor (Theorem 1).

(ii) For $d = 2$ and $k \in \{2, 3\}$, the velocity error satisfies

$$\|u(t_m) - u_h^m\|_0 \le C\big((\nu + h)^{1/2}h^k + \Delta t^2\big),$$

where C depends neither on the pressure nor on ν^{-1} (Theorem 2). For $\nu \lesssim h$ this is $O(h^{k+1/2} + \Delta t^2)$, the fully discrete counterpart of [16, Thm. 3.7]. In three dimensions the same estimate

holds provided a potential-recovery property is available, which, as far as we know, has not been proved yet.

(iii) The normalized stabilization needs the extrapolated transport field to stay bounded from above and below. We obtain this from a preliminary robust estimate under the mild condition $\Delta t \le c_0 h^{1/2+\sigma}$ (Corollary 1); with a clamped normalization, the condition is no longer needed for the velocity estimate (Remark 4).

(iv) For the pressure we prove a robust $\ell^2(L^2)$ bound of order $h^{k-1} + \Delta t^2/h$ (Theorem 3), and we indicate how the optimal rate could be reached (Remark 5).

Compared with [16], the analysis needs two new ingredients. One is an estimate for the consistency error of the stabilization that is of order h^{2k+1} uniformly in ν and does not feel the extrapolation defect $2u^n - u^{n-1} - u^{n+1}$ (Lemma 2). The other is an induction over the time levels that keeps the extrapolated transport field bounded from above and below (Lemma 3 and Corollary 1). The table below places our velocity estimate among related fully discrete, ν-robust results (k: velocity degree; ℓ: degree in time).

Reference	Space / time	Velocity bound	Time-step condition
[14] (grad-div)	inf-sup stable / BDF-q	$h^k + \Delta t^q$	$\Delta t\, h^{-2} \le C$, $h^{-d/2}\Delta t^q \le C$
[5]	SUPG-curl, div-free / 2nd order	$\tau^2 + h^{k+1/2}$ ($\nu \lesssim h$)	$\tau \lesssim h^{(s+2)/2}$ (asymptotic, $s \le 4$)
[6]	$H(\mathrm{div})$-DG / space–time DG	$\tau^{\ell+1} + \sqrt{\nu + h}\, h^k$	$\tau \le C$
this paper	SV + CIP / BDF2–IMEX	$\Delta t^2 + \sqrt{\nu + h}\, h^k$	$\Delta t \le c_0 h^{1/2+\sigma}$ ($d = 2$); none with clamping (Rem. 4)

Table, row [GAJN25]: check the condition $\Delta t\, h^{-2} \le C$ (it means $\Delta t \lesssim h^2$).

In the numerical part we test the Scott–Vogelius discretization, with and without the CIP stabilization, on a Taylor–Green vortex with manufactured forcing, on a three-dimensional travelling vortex, and on an under-resolved double shear layer; a Taylor–Hood discretization serves as a non-divergence-free reference in the pressure-robustness tests and where it is outperformed. The experiments measure the spatial rates predicted by the theory, confirm second-order accuracy in time, demonstrate the robustness with respect to ν and the pressure robustness of the divergence-free discretization, and show how the CIP terms suppress spurious oscillations.

Outline. Section 2 introduces the model problem and the spatially discrete CIP method. In Section 3 we formulate the BDF2–IMEX scheme, prove its energy stability, and derive the velocity error estimate. The pressure is treated in Section 4, and the numerical studies follow in Section 5. Appendix A collects the estimate for the discrete time derivative used in the pressure bound.

2 Model problem and spatial discretization

2.1 Model problem

Let $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$, be a bounded polyhedral Lipschitz domain, and let $T > 0$. Given a viscosity $\nu > 0$, a body force $f : (0, T] \times \Omega \to \mathbb{R}^d$, and an initial velocity u_0, the unknowns are the velocity field $u : (0, T] \times \Omega \to \mathbb{R}^d$ and the kinematic pressure $p : (0, T] \times \Omega \to \mathbb{R}$. The strong form of the

model problem is

$$\begin{aligned} \partial_t u - \nu\Delta u + (u\cdot\nabla)u + \nabla p &= f && \text{in } (0,T]\times\Omega,\\ \nabla\cdot u &= 0 && \text{in } (0,T]\times\Omega,\\ u &= 0 && \text{on } [0,T]\times\partial\Omega,\\ u(0) &= u_0 && \text{in } \Omega. \end{aligned} \tag{1}$$

The homogeneous Dirichlet condition models a no-slip wall.

For the weak formulation, define

$$V = H_0^1(\Omega)^d, \qquad Q = L_0^2(\Omega) = \Big\{ q \in L^2(\Omega) : \int_\Omega q\,\mathrm{d}x = 0 \Big\}.$$

Find $(u(t), p(t)) \in V \times Q$, with $u(0) = u_0$, such that for all $(v, q) \in V \times Q$ and for almost every $t \in (0, T]$,

$$\begin{aligned} (\partial_t u, v) + \nu(\nabla u, \nabla v) + ((u\cdot\nabla)u, v) - (p, \nabla\cdot v) &= (f, v),\\ (\nabla\cdot u, q) &= 0. \end{aligned} \tag{2}$$

Here $(\cdot,\cdot)$ is the $L^2(\Omega)$ inner product, and the pressure is sought in $L_0^2(\Omega)$ because it is determined only up to a constant. Testing only with divergence-free functions from

$$V_{\mathrm{div}} = \{ v \in H_0^1(\Omega)^d : \nabla\cdot v = 0 \},$$

removes the pressure, and the velocity satisfies

$$(\partial_t u, v) + \nu(\nabla u, \nabla v) + ((u\cdot\nabla)u, v) = (f, v) \qquad \forall v \in V_{\mathrm{div}}. \tag{3}$$

Notation. For a function v of time we abbreviate $v^n := v(t_n)$, so that $u^n = u(t_n)$, $p^n = p(t_n)$ and $f^n = f(t_n)$. We write $\|\cdot\|_0$ for the $L^2(\Omega)$ norm and $\|\cdot\|_{0,\infty}$ for the $L^\infty(\Omega)$ norm. For piecewise smooth functions, $(\cdot,\cdot)_h$ and $\|\cdot\|_{0,h}$ denote the elementwise (broken) L^2 inner product and norm over the triangulation, and $\|D^j v\|_{0,h}$ is the broken L^2 norm of all j-th order derivatives. For a family $\mathcal{F}$ of faces, $\|\varphi\|_{0,\mathcal{F}}^2 = \sum_{F\in\mathcal{F}} \|\varphi\|_{0,F}^2$, and face-wise weights are written inside the norm, as in $\|h\varphi\|_{0,\mathcal{F}}$; since the meshes are quasi-uniform, we do not distinguish between the face size h_F and h in the estimates. Finally, C denotes a generic constant that is independent of h, Δt, of $\nu \le \nu_{\max}$ and of the pressure, and may change from one occurrence to the next; dependencies that matter are stated explicitly.

2.2 Spatial discretization and stabilization

Let $\mathcal{T}_h$ be a shape-regular, quasi-uniform triangulation of Ω, with mesh size h. We choose conforming finite element spaces $V_h \subset H_0^1(\Omega)^d$ and $Q_h \subset L_0^2(\Omega)$ that satisfy the discrete inf-sup condition: there exists a constant $\beta > 0$, independent of h, such that

$$\inf_{q_h\in Q_h} \sup_{v_h\in V_h} \frac{(q_h, \nabla\cdot v_h)}{\|q_h\|_0 \|\nabla v_h\|_0} \ge \beta.$$

Following the divergence-free setting from [3, 16], we also assume

$$\nabla\cdot V_h \subset Q_h, \qquad V_h^{\mathrm{div}} = \{ v_h \in V_h : \nabla\cdot v_h = 0 \}.$$

A standard example is the Scott–Vogelius pair on barycentrically refined triangulations (Alfeld splits of a shape-regular, quasi-uniform triangulation),

$$V_h = \{v_h \in H_0^1(\Omega)^d : v_h|_K \in P_k(K)^d \ \forall K \in \mathcal{T}_h\},$$
$$Q_h = \{q_h \in L_0^2(\Omega) : q_h|_K \in P_{k-1}(K) \ \forall K \in \mathcal{T}_h\}.$$

for suitable polynomial degree $k \geq d$ (so $k \geq 2$ if $d = 2$, $k \geq 3$ if $d = 3$); see [2] for $d = 2$, [23] for $d = 3$ and [18] for general d. This choice gives pointwise divergence-free discrete velocities and hence pressure-robust velocity errors. When this pair is used, $\mathcal{T}_h$ stands for the refined (barycentric) triangulation itself; in particular, the set $\mathcal{F}_h^i$ of interior faces introduced below consists of the faces of the refined mesh, including those created by the splits, and h is its mesh size.

The semi-discrete problem reads: find $(u_h(t), p_h(t)) \in V_h \times Q_h$, with a suitable approximation $u_h(0) \approx u_0$, $u_h(0) \in V_h^{\mathrm{div}}$, such that for all $(v_h, q_h) \in V_h \times Q_h$,

$$\begin{aligned}(\partial_t u_h, v_h) + \nu(\nabla u_h, \nabla v_h) + ((u_h \cdot \nabla)u_h, v_h) - (p_h, \nabla \cdot v_h) &= (f, v_h),\\ (\nabla \cdot u_h, q_h) &= 0.\end{aligned} \tag{4}$$

Following the stabilized divergence-free approach in [3], extended to the Navier–Stokes equations in [16], we introduce a continuous interior penalty (CIP) stabilization that penalizes jumps of convective derivatives over the interior mesh faces. Let $\mathcal{F}_h^i$ denote the set of interior faces. For each $F = \partial K^+ \cap \partial K^- \in \mathcal{F}_h^i$, fix a unit normal n_F directed from K^+ to K^-. If $w^\pm$ denotes the trace on F from $K^\pm$, define the jump componentwise by

$$[\![w]\!]_F := w^+ - w^-.$$

In the following, we write $n = n_F$ and suppress the subscript F on the jump. On each element K, define the elementwise operator

$$(Bw)|_K = \nabla \times ((u_h \cdot \nabla)w)|_K.$$

For non-dimensional stabilization parameters $\delta_1, \delta_2, \delta_3 > 0$, the stabilization term S is defined by

$$S(u_h, v_h) = \|u_h\|_{0,\infty}^{-1} \sum_{j=1}^{3} \delta_j S_j(u_h, v_h) \quad (\text{with } S := 0 \text{ if } \|u_h\|_{0,\infty} = 0),$$

where

$$S_1(u_h, v_h) = \sum_{F \in \mathcal{F}_h^i} \int_F h_F^2 \, [\![(u_h \cdot \nabla)u_h \times n]\!] \cdot [\![(u_h \cdot \nabla)v_h \times n]\!] \, \mathrm{d}s,$$
$$S_2(u_h, v_h) = \sum_{F \in \mathcal{F}_h^i} \int_F h_F^4 \, [\![Bu_h]\!] \cdot [\![Bv_h]\!] \, \mathrm{d}s,$$
$$S_3(u_h, v_h) = \sum_{F \in \mathcal{F}_h^i} \int_F h_F^6 \, [\![\nabla Bu_h]\!] : [\![\nabla Bv_h]\!] \, \mathrm{d}s.$$

Thus the CIP-stabilized semidiscrete problem is: find $(u_h(t), p_h(t)) \in V_h \times Q_h$ such that

$$\begin{aligned}(\partial_t u_h, v_h) + \nu(\nabla u_h, \nabla v_h) + ((u_h \cdot \nabla)u_h, v_h) + S(u_h, v_h) - (p_h, \nabla \cdot v_h) &= (f, v_h),\\ (\nabla \cdot u_h, q_h) &= 0,\end{aligned} \tag{5}$$

for all $(v_h, q_h) \in V_h \times Q_h$. Compared with (4), the role of S in (5) is to add streamline dissipation through interelement jump terms while preserving the pressure-robust structure of the divergence-free finite element method. By trace and inverse inequalities, for a transport field $\tilde{u} \in V_h$ and $v_h \in V_h$ one has

$$S_j(v_h, v_h) \le C\, h\, \|\tilde{u}\|_{0,\infty}^2 \|\nabla v_h\|_0^2 \tag{6}$$

with C depending on the trace and inverse-inequality constants (for S_2, S_3 through $\|\nabla^m \tilde{u}\|_{0,\infty} \le Ch^{-m}\|\tilde{u}\|_{0,\infty}$, $m = 1, 2$) and the shape regularity. This is a bound on the *raw*, unnormalized form S_j as defined above; two powers of $\|\tilde{u}\|_{0,\infty}$ appear because $\tilde{u}$ occurs twice inside the jump operators (S_1 directly, S_2, S_3 through B). Dividing (6) by $\|\tilde{u}\|_{0,\infty}$, as the scheme does, gives the normalized bound actually used below and in Section 4:

$$\|\tilde{u}\|_{0,\infty}^{-1} S_j(v_h, v_h) \le C\, h\, \|\tilde{u}\|_{0,\infty} \|\nabla v_h\|_0^2, \qquad \text{hence} \qquad S_{\mathrm{IMEX}}^{n+1}(v_h, v_h) \le C\, h\, \|\tilde{u}_h^{n+1}\|_{0,\infty} \|\nabla v_h\|_0^2. \tag{7}$$

The normalized estimate (7) is an upper bound and is used in the stabilization-residual estimate in the pressure-error proof. The stabilization-consistency estimate (Lemma 2) divides face terms that are *not* proportional to $\|\tilde{u}\|_{0,\infty}^2$ by $\|\tilde{u}\|_{0,\infty}$ and therefore needs a *lower* bound on $\|\tilde{u}\|_{0,\infty}$, see Assumption 1 and [16, (48)]. As suggested in [16, Remark 3.4], this can also be enforced in the scheme by replacing $\|\tilde{u}\|_{0,\infty}^{-1}$ with $\max(u_{\min}, \|\tilde{u}\|_{0,\infty})^{-1}$ for a fixed $u_{\min} > 0$, the proofs below apply to that variant with only notational changes; a two-sided clamp, which also removes the need for an upper bound, is discussed in Remark 4.

3 BDF2–IMEX time discretizationand velocity error analysis

We now discretize the stabilized semidiscrete problem (5) in time. Let $0 = t_0 < t_1 < \cdots < t_N = T$ be a uniform partition of $[0, T]$, with time step $\Delta t = t_n - t_{n-1}$. For a sequence $\{w^n\}_{n=0}^N$, the BDF2 approximation of the time derivative at t_{n+1} is

$$D_t^{\mathrm{BDF2}} w^{n+1} := \frac{3w^{n+1} - 4w^n + w^{n-1}}{2\Delta t}.$$

To obtain a linear problem at each time step, the nonlinear convective velocity is treated explicitly by the second-order extrapolation

$$\tilde{u}_h^{n+1} = 2u_h^n - u_h^{n-1}.$$

The viscous term, pressure term, incompressibility constraint, and test-function factor in the convection operator are treated implicitly. This gives an implicit–explicit (IMEX) BDF2 method. Given (u_h^n, p_h^n) and (u_h^{n-1}, p_h^{n-1}), find $(u_h^{n+1}, p_h^{n+1}) \in V_h \times Q_h$ such that for all $(v_h, q_h) \in V_h \times Q_h$,

$$\begin{aligned} (D_t^{\mathrm{BDF2}} u_h^{n+1}, v_h) + \nu(\nabla u_h^{n+1}, \nabla v_h) + ((\tilde{u}_h^{n+1} \cdot \nabla) u_h^{n+1}, v_h) + S_{\mathrm{IMEX}}^{n+1}(u_h^{n+1}, v_h) \\ -(p_h^{n+1}, \nabla \cdot v_h) = (f^{n+1}, v_h), \\ (\nabla \cdot u_h^{n+1}, q_h) = 0. \end{aligned} \tag{8}$$

Here $f^{n+1} = f(t_{n+1})$. The IMEX version of the CIP stabilization is formed with the same extrapolated transport field. On each element K, define

$$(B_{\mathrm{IMEX}}^{n+1} w)|_K = \nabla \times ((\tilde{u}_h^{n+1} \cdot \nabla) w)|_K.$$

Then

$$S_{\mathrm{IMEX}}^{n+1}(u_h^{n+1}, v_h) = \|\tilde{u}_h^{n+1}\|_{0,\infty}^{-1} \sum_{j=1}^{3} \delta_j S_{j,\mathrm{IMEX}}^{n+1}(u_h^{n+1}, v_h),$$

with the convention $S^{n+1}_{\mathrm{IMEX}} := 0$ whenever $\|\tilde{u}_h^{n+1}\|_{0,\infty} = 0$, and

$$S^{n+1}_{1,\mathrm{IMEX}}(u_h^{n+1}, v_h) = \sum_F \int_F h_F^2 [\![(\tilde{u}_h^{n+1}\cdot\nabla)u_h^{n+1}\times n]\!]\cdot[\![(\tilde{u}_h^{n+1}\cdot\nabla)v_h\times n]\!]\mathrm{d}s,$$

$$S^{n+1}_{2,\mathrm{IMEX}}(u_h^{n+1}, v_h) = \sum_F \int_F h_F^4 [\![B^{n+1}_{\mathrm{IMEX}}u_h^{n+1}]\!]\cdot[\![B^{n+1}_{\mathrm{IMEX}}v_h]\!]\mathrm{d}s,$$

$$S^{n+1}_{3,\mathrm{IMEX}}(u_h^{n+1}, v_h) = \sum_F \int_F h_F^6 [\![\nabla B^{n+1}_{\mathrm{IMEX}}u_h^{n+1}]\!]:[\![\nabla B^{n+1}_{\mathrm{IMEX}}v_h]\!]\mathrm{d}s.$$

Thus, at each time level, the convective coefficients and stabilization weights are known from previous time levels, while u_h^{n+1} and p_h^{n+1} are computed implicitly. Since BDF2 is a two-step method, the first value u_h^1 can be obtained by a first-order IMEX Euler step or by another consistent one-step starter.

3.1 Stability of the fully discrete scheme

We now state a basic energy stability estimate for the fully discrete BDF2–IMEX scheme (8). For notational convenience, define

$$c(w;u,v) = ((w\cdot\nabla)u, v), \qquad \tilde{u}_h^{n+1} = 2u_h^n - u_h^{n-1}.$$

Since the scheme enforces $(\nabla\cdot u_h^j, q_h) = 0$ for all $q_h \in Q_h$, the extrapolated velocity $\tilde{u}_h^{n+1}$ is also discretely divergence-free *provided the starting value* u_h^0 *is also in* V_h^{div} (e.g. the modified Stokes projection of u_0; a nodal interpolant of u_0 is not divergence-free). In the exactly divergence-free Scott–Vogelius setting, this implies $\nabla\cdot\tilde{u}_h^{n+1} = 0$ pointwise.

Theorem 1. *Assume that the velocity space is exactly divergence-free, that the no-slip boundary condition is imposed strongly, that* $u_h^0, u_h^1 \in V_h^{\mathrm{div}}$ *(so that* $\nabla\cdot\tilde{u}_h^{n+1} = 0$ *for all* $n \geq 1$*), and that* $f^{n+1} \in L^2(\Omega)^d$*. Then, for every* $\Delta t > 0$ *and every* $m \geq 2$*, the solution of* (8) *satisfies*

$$\|u_h^m\|_0^2 + \|2u_h^m - u_h^{m-1}\|_0^2 + 4\nu\Delta t\sum_{n=1}^{m-1}\|\nabla u_h^{n+1}\|_0^2 + 4\Delta t\sum_{n=1}^{m-1} S^{n+1}_{\mathrm{IMEX}}(u_h^{n+1}, u_h^{n+1})$$
$$\leq 3\big(\|u_h^1\|_0^2 + \|2u_h^1 - u_h^0\|_0^2\big) + 18\Big(\Delta t\sum_{n=1}^{m-1}\|f^{n+1}\|_0\Big)^2. \quad (9)$$

The bound holds without restriction on Δt*, contains no Gronwall factor, and is independent of* ν*.*

Proof. Choose $v_h = u_h^{n+1}$ and $q_h = p_h^{n+1}$ in (8); the pressure term vanishes because $(p_h^{n+1}, \nabla\cdot u_h^{n+1}) = 0$. By the G-stability identity of BDF2 (see, e.g., [30, Sec. V.6] or [15])

$$(3a - 4b + c, a) = \frac{1}{2}\big(\|a\|_0^2 + \|2a - b\|_0^2 - \|b\|_0^2 - \|2b - c\|_0^2 + \|a - 2b + c\|_0^2\big),$$

with $a = u_h^{n+1}$, $b = u_h^n$, $c = u_h^{n-1}$, one has

$$(D_t^{\mathrm{BDF2}}u_h^{n+1}, u_h^{n+1}) = \frac{1}{4\Delta t}\big(F^{n+1} - F^n + D^{n+1}\big), \qquad F^{n+1} = \|u_h^{n+1}\|_0^2 + \|2u_h^{n+1} - u_h^n\|_0^2,$$

with $D^{n+1} = \|u_h^{n+1} - 2u_h^n + u_h^{n-1}\|_0^2 \geq 0$. The CIP term is nonnegative, and $c(\tilde{u}_h^{n+1}; u_h^{n+1}, u_h^{n+1}) = 0$ because $\nabla \cdot \tilde{u}_h^{n+1} = 0$ and $u_h^{n+1} \in H_0^1(\Omega)^d$ [29, 31]. Bounding $(f^{n+1}, u_h^{n+1}) \leq \|f^{n+1}\|_0 (F^{n+1})^{1/2}$ and multiplying by $4\Delta t$ gives

$$F^{n+1} - F^n + 4\Delta t\big(\nu\|\nabla u_h^{n+1}\|_0^2 + S_{\mathrm{IMEX}}^{n+1}(u_h^{n+1}, u_h^{n+1})\big) \leq 4\Delta t\, \|f^{n+1}\|_0\, (F^{n+1})^{1/2}.$$

Sum from $n = 1$ to $m-1$ and write $A_m := \Delta t \sum_{n=1}^{m-1} \|f^{n+1}\|_0$ and $M_m := \max_{1\leq j\leq m} F^j$:

$$F^m + 4\Delta t \sum_{n=1}^{m-1} \big(\nu\|\nabla u_h^{n+1}\|_0^2 + S_{\mathrm{IMEX}}^{n+1}(u_h^{n+1}, u_h^{n+1})\big) \leq F^1 + 4A_m M_m^{1/2}. \tag{10}$$

Applying (10) at an index $j \leq m$ with $F^j = M_m$ and using $A_j \leq A_m$ gives $M_m \leq F^1 + 4A_m M_m^{1/2}$, hence $M_m^{1/2} \leq 2A_m + (4A_m^2 + F^1)^{1/2} \leq (F^1)^{1/2} + 4A_m$. Inserting this into (10) and using $4A_m(F^1)^{1/2} \leq 2F^1 + 2A_m^2$ proves (9). □

In the following, the divergence-free structure is used to remove pressure from the velocity estimate, while the CIP terms control convective derivatives through interelement jumps.

3.2 Approximation assumptions and projections

Assume that the exact solution is sufficiently regular, for example

$$u \in L^\infty(0,T; H^{k+1}(\Omega)^d), \qquad \partial_t u \in L^2(0,T; H^{k+1}(\Omega)^d),$$

and, for the viscosity-independent treatment of the temporal residuals, assume explicitly that

$$\partial_{tt} u \in L^\infty(0,T; L^2(\Omega)^d), \qquad \partial_{ttt} u \in L^2(0,T; L^2(\Omega)^d).$$

For $d = 2$ and Ω simply connected, a divergence-free $u \in H_0^1(\Omega)^2$ has a stream function $z \in H_0^2(\Omega)$ with $\nabla \times z = u$, unique up to a constant; since ∇z is a rotation of u, $\|z\|_{H^{r+1}} \leq C\|u\|_{H^r}$ holds without any elliptic regularity. For $d = 3$ one needs, in addition, a vector potential with boundary conditions that put $\nabla \times P_h z$ in $H_0^1(\Omega)^3$, together with the corresponding regularity; see [29, 11]. Following [3, 16], we use the following structural assumptions:

1. $\nabla \cdot V_h \subset Q_h$, so the discrete velocity can be exactly divergence-free.
2. The pair (V_h, Q_h) is inf-sup stable.
3. There exists a potential space $Z_h \subset Z$ such that $\nabla \times Z_h = V_h^{\mathrm{div}}$.
4. If $P_h : Z \to Z_h$ is the L^2-projection, then for $|\alpha| \leq \min(5, k+2)$,

   $$\|h^{|\alpha|}\partial^\alpha (z - P_h z)\|_{0,h} \leq C h^{k+2} \|z\|_{H^{k+2}}.$$

   In [3, 16] this is assumed for $|\alpha| \leq k+1$; the case $|\alpha| = k+2$ is immediate, since $P_h z$ is piecewise polynomial of degree $k+1$, so that $\partial^\alpha P_h z = 0$ elementwise. For $d = 2$ and Alfeld splits it is proved in [3, Sec. 4] for quasi-uniform meshes.
5. The potential space has the following recovery property. For every broken polynomial χ_h in the range of the elementwise operator used below,

   $$\inf_{m_h \in Z_h} \|h^{3/2}(\chi_h - m_h)\|_{0,h} \leq C\left(\|h^2 [\![\chi_h]\!]\|_{0,\mathcal{F}_h^i} + \|h^3 [\![\nabla \chi_h]\!]\|_{0,\mathcal{F}_h^i}\right). \tag{11}$$

This is the discrete potential-recovery assumption corresponding to the two additional penalties S_2 and S_3; it is the step that converts their face jumps into control of a broken volume term. For Scott–Vogelius elements on Alfeld (Clough–Tocher) splits, with Z_h the C^1 Clough–Tocher space of degree $k+1$, (11) is proved in [3, Lemma 4.1] for $d = 2$ and for broken polynomials χ_h of degree $\leq k+1$ on the refined mesh. For $d = 3$ (potential spaces from [13]) it is, to our knowledge, not proved; [3, Sec. 6] states that the proof can be extended.

Scope of the error analysis. From here on we restrict the error analysis to $d = 2$ and $k \in \{2, 3\}$. The reason is the recovery step: for these degrees, with $\beta = \tilde{u}_h^{n+1}$ and $e \in V_h$, the function $\nabla \times ((\beta \cdot \nabla)e)$ is elementwise a polynomial of degree $2k-2 \leq k+1$, so (11) can be applied to it directly, without an auxiliary low-order transport field (Lemma 4). Higher degrees could be handled by inserting a piecewise constant Raviart–Thomas approximation of the transport field as in [3], at the cost of an additional a priori L^2 bound on the discrete error; we do not pursue this here. In three dimensions, where $k = 3$, everything below carries over as soon as (11) is available for the potential spaces of [13] and assumption 4 holds for the associated L^2-projection. Define the divergence-free projected velocity

$$w_h(t) = \nabla \times P_h z(t) \in V_h^{\mathrm{div}}.$$

Then, for smooth solutions,

$$\|u(t) - w_h(t)\|_0 + h\|\nabla(u(t) - w_h(t))\|_0 \leq Ch^{k+1}\|u(t)\|_{H^{k+1}}.$$

Use the modified Stokes projection $s_h \in V_h^{\mathrm{div}}$, characterized by

$$(\nabla s_h, \nabla \varphi_h) = (\nabla u, \nabla \varphi_h) \qquad \forall \varphi_h \in V_h^{\mathrm{div}},$$

which satisfies

$$\|s_h - u\|_0 + h\|\nabla(s_h - u)\|_0 \leq Ch^{k+1}\|u\|_{H^{k+1}}.$$

The pointwise bounds

$$\|w_h\|_{0,\infty} \leq C\|u\|_{0,\infty} + Ch^{k+1-d/2}\|z\|_{H^{k+2}}, \qquad \|\nabla w_h\|_{0,\infty} \leq C\|\nabla u\|_{0,\infty} + Ch^{k-d/2}\|z\|_{H^{k+2}} \tag{12}$$

are used to keep the convective constants independent of inverse powers of the viscosity.

At each time level, write

$$u(t_n) - u_h^n = \eta^n - e_h^n, \qquad \eta^n = u(t_n) - w_h^n, \qquad e_h^n = u_h^n - w_h^n.$$

The approximation part η^n is controlled by the projection estimates, while the discrete part e_h^n is controlled by an energy argument. Since both u_h^n and w_h^n are discretely divergence-free, we have $e_h^n \in V_h^{\mathrm{div}}$. This is the key pressure-robustness mechanism: testing the error equation with e_h^{n+1} eliminates the pressure contribution.

For the BDF2 time derivative, define the local truncation residual

$$\tau^{n+1} = D_t^{\mathrm{BDF2}} u(t_{n+1}) - \partial_t u(t_{n+1}).$$

If u is sufficiently smooth in time, then

$$\Delta t \sum_{n=1}^{N-1} \|\tau^{n+1}\|_0^2 \leq C\Delta t^4 \|\partial_{ttt} u\|^2_{L^2(0,T;L^2)}.$$

We also use the projection time residual $\rho^{n+1} = D_t^{\mathrm{BDF2}}\eta^{n+1}$, which satisfies (both bounds are standard; see, e.g., [15])

$$\Delta t \sum_{n=1}^{N-1} \|\rho^{n+1}\|_0^2 \le Ch^{2k+2}\|\partial_t u\|^2_{L^2(0,T;H^{k+1})}.$$

The nonlinear consistency analysis below requires two-sided control of $\|\tilde{u}_h^{n+1}\|_{0,\infty}$: a lower bound in Lemma 2 and an upper bound in Lemma 4. This control is derived in Corollary 1 from a preliminary robust estimate, in the spirit of [16, Prop. 3.3]. For $k \in \{2,3\}$, no a priori bound of the form $\|e_h^n\|_0 \le \Gamma h^2$ (cf. [16, Lemma 3.6]) is needed.

Subtracting the fully discrete scheme (8) from the exact weak equation (2) tested with $v_h \in V_h^{\mathrm{div}}$ and inserting the splitting above gives

$$\begin{aligned}(D_t^{\mathrm{BDF2}}e_h^{n+1}, v_h) + \nu(\nabla e_h^{n+1}, \nabla v_h) + S^{n+1}_{\mathrm{IMEX}}(e_h^{n+1}, v_h) = &- (\tau^{n+1}, v_h) + (\rho^{n+1}, v_h)\\ &+ \nu(\nabla\eta^{n+1}, \nabla v_h) + N^{n+1}(v_h) + S^{n+1}(v_h),\end{aligned} \tag{13}$$

where the nonlinear residual is

$$N^{n+1}(v_h) = \big((u(t_{n+1})\cdot\nabla)u(t_{n+1}), v_h\big) - \big((\tilde{u}_h^{n+1}\cdot\nabla)u_h^{n+1}, v_h\big),$$

and the stabilization consistency residual is

$$S^{n+1}(v_h) = -S^{n+1}_{\mathrm{IMEX}}(w_h^{n+1}, v_h).$$

The pressure does not appear in this equation. For later use, define the mesh-dependent potential norm

$$\|\xi\|_*^2 = h \sum_{j=0}^{\min(5,k+2)} h^{2j-4}\|D^j\xi\|^2_{0,h}, \tag{14}$$

and set $\varepsilon_h := (\nu + h)h^{2k}$ (the problem is non-dimensional, so the characteristic velocity is one). The norm carries no viscous weight: the ν-dependent part of the error stems only from the viscous projection term, see Remark 3(i). We first state the lower bound needed by the normalized stabilization.

Assumption 1. *There is $u_{\min} > 0$ such that*

$$\|u(t)\|_{0,\infty} \ge 2u_{\min} \quad \textit{for all} \quad t \in [0,T].$$

(If the normalization in the scheme is replaced by $\max(u_{\min}, \|\tilde{u}_h^{n+1}\|_{0,\infty})^{-1}$, this assumption can be dropped; see also Remark 4.) In addition,

$$\partial_{tt}u \in L^\infty(0,T;L^\infty(\Omega)^d) \quad \textit{and} \quad u \in L^\infty(0,T;W^{2,\infty}(\Omega)^d).$$

Weakened from $\partial_{tt}u \in W^{2,\infty}$, $u \in W^{3,\infty}$: these appear unused; please confirm.

3.3 Auxiliary estimates

Throughout, $E^m = \|e_h^m\|_0^2 + \|2e_h^m - e_h^{m-1}\|_0^2$ for $m \ge 1$, and $E^0 := \|e_h^0\|_0^2$. We assume $u_h^0, u_h^1 \in V_h^{\mathrm{div}}$, so that $\tilde{u}_h^{n+1} \in V_h^{\mathrm{div}}$ for all $n \ge 1$. The following elementary bounds do not use the recovery property (11).

Lemma 1 (Elementary estimates)**.** *Assume the regularity and approximation hypotheses of Section 3.2 and $k \geq d$. For $1 \leq n \leq N-1$ the following estimates hold, with constants independent of h, Δt, n, $\nu \leq \nu_{\max}$ and the pressure.*

(i) (BDF2 energy)

$$(D_t^{\mathrm{BDF2}} e_h^{n+1}, e_h^{n+1}) \geq \frac{1}{4\Delta t}\big(E^{n+1} - E^n\big). \tag{15}$$

(ii) (Temporal residuals)

$$|(\tau^{n+1}, e_h^{n+1})| + |(\rho^{n+1}, e_h^{n+1})| \leq \frac{1}{2}\big(\|\tau^{n+1}\|_0^2 + \|\rho^{n+1}\|_0^2\big) + E^{n+1}. \tag{16}$$

(iii) (Viscous projection term)

$$\nu|(\nabla\eta^{n+1}, \nabla e_h^{n+1})| \leq \frac{\nu}{4}\|\nabla e_h^{n+1}\|_0^2 + C\nu\|\nabla\eta^{n+1}\|_0^2, \qquad \|\nabla\eta^{n+1}\|_0 \leq Ch^k\|u(t_{n+1})\|_{H^{k+1}}. \tag{17}$$

(iv) (Potential projection)

$$\|z^{n+1} - P_h z^{n+1}\|_*^2 \leq Ch^{2k+1} \leq C\varepsilon_h. \tag{18}$$

(v) (Nonlinear residual, first part) The nonlinear residual satisfies

$$N^{n+1}(e_h^{n+1}) = ((u^{n+1} - \tilde{u}_h^{n+1}) \cdot \nabla u^{n+1}, e_h^{n+1}) + ((\tilde{u}_h^{n+1} \cdot \nabla)\eta^{n+1}, e_h^{n+1})$$

and

$$\big|N^{n+1}(e_h^{n+1}) - ((\tilde{u}_h^{n+1} \cdot \nabla)\eta^{n+1}, e_h^{n+1})\big| \leq C\big(E^{n+1} + E^n\big) + C\big(\Delta t^4 + h^{2k+2}\big). \tag{19}$$

(vi) (Crude transport bound)

$$|((\tilde{u}_h^{n+1} \cdot \nabla)\eta^{n+1}, e_h^{n+1})| \leq C\,h^{2k} + C\,(E^{n+1} + E^n). \tag{20}$$

Proof. (i) follows from the G-stability identity used in the proof of Theorem 1, applied to e_h, after dropping $\|e_h^{n+1} - 2e_h^n + e_h^{n-1}\|_0^2 \geq 0$. (ii) and (iii) follow from the Cauchy–Schwarz and Young inequalities, $\|e_h^{n+1}\|_0^2 \leq E^{n+1}$ and the approximation properties of w_h. (iv) By assumption 4 of Section 3.2, $h^{2j-4}\|D^j(z - P_h z)\|_{0,h}^2 \leq Ch^{2k}$ for $0 \leq j \leq \min(5, k+2)$; summing over j and multiplying by h gives (18).

(v) First we write

$$u^{n+1} - u_h^{n+1} = \eta^{n+1} - e_h^{n+1}.$$

Since $\tilde{u}_h^{n+1} \in V_h^{\mathrm{div}} \subset H_0^1(\Omega)^d$ is divergence-free, skew-symmetry [29, 31] gives

$$((\tilde{u}_h^{n+1} \cdot \nabla)e_h^{n+1}, e_h^{n+1}) = 0,$$

which yields the stated decomposition of $N^{n+1}(e_h^{n+1})$. Moreover,

$$u^{n+1} - \tilde{u}_h^{n+1} = (u^{n+1} - 2u^n + u^{n-1}) + (2\eta^n - \eta^{n-1}) - (2e_h^n - e_h^{n-1}). \tag{21}$$

Here

$$\|u^{n+1} - 2u^n + u^{n-1}\|_0 \leq C\Delta t^2\|\partial_{tt}u\|_{L^\infty(t_{n-1},t_{n+1};L^2)},$$

$$\|2\eta^n - \eta^{n-1}\|_0 \leq Ch^{k+1}, \quad \text{and} \quad \|2e_h^n - e_h^{n-1}\|_0^2 \leq E^n.$$

Hölder's and Young's inequalities, with $\|\nabla u^{n+1}\|_{0,\infty} \leq C$, give (19).

(vi) Write

$$\tilde{u}_h^{n+1} = \tilde{w}^{n+1} + \tilde{e}^{\,n+1} \quad \text{with} \quad \tilde{w}^{n+1} = 2w_h^n - w_h^{n-1} \text{ and } \tilde{e}^{\,n+1} = 2e_h^n - e_h^{n-1}.$$

By the pointwise bounds on w_h in Section 3.2 and by (iii),

$$|((\tilde{w}^{n+1}\cdot\nabla)\eta^{n+1}, e_h^{n+1})| \le 3\max_j \|w_h^j\|_{0,\infty}\|\nabla\eta^{n+1}\|_0\|e_h^{n+1}\|_0 \le Ch^{2k} + CE^{n+1}.$$

Since $k \ge d$, the pointwise bounds give

$$\|\nabla\eta^{n+1}\|_{0,\infty} \le \|\nabla u^{n+1}\|_{0,\infty} + \|\nabla w_h^{n+1}\|_{0,\infty} \le C.$$

Hence

$$|((\tilde{e}^{\,n+1}\cdot\nabla)\eta^{n+1}, e_h^{n+1})| \le C\|\tilde{e}^{\,n+1}\|_0\|e_h^{n+1}\|_0 \le C(E^n + E^{n+1}).$$

Adding the two bounds proves (20). □

The next lemma bounds the consistency error of the stabilization. It uses only the lower bound on the transport field, and not (11).

Lemma 2 (Stabilization consistency)**.** *Let $k \ge d$, let $u \in L^\infty(0,T;W^{2,\infty}(\Omega)^d)$ satisfy the hypotheses of Section 3.2, and suppose that $\|\tilde{u}_h^{n+1}\|_{0,\infty} \ge u_{\min}$. Then, with $\xi = z^{n+1} - P_h z^{n+1}$,*

$$S_{\mathrm{IMEX}}^{n+1}(w_h^{n+1}, w_h^{n+1}) \le C_S\Big(h\,E^n + h^{2k+3} + \|\xi\|_*^2\Big) \le C_S\Big(h\,E^n + h^{2k+1}\Big). \tag{22}$$

The constant C_S depends only on $\delta_{\max}$, $u_{\min}^{-1}$, $\|u\|_{L^\infty(0,T;W^{2,\infty})}$, $C_\nabla := \max_j \|\nabla w_h^j\|_{0,\infty}$, and the trace, inverse and approximation constants. It is independent of h, Δt, n, ν and the pressure. In particular, the consistency error of the stabilization is $O(h^{2k+1})$ uniformly in ν.

Proof. Put $\beta = \tilde{u}_h^{n+1}$, $w = w_h^{n+1}$ and $\beta_* = 2u^n - u^{n-1}$. From $u_h^j = u^j - \eta^j + e_h^j$,

$$\beta = \beta_* + r, \qquad r = \rho_e - \rho_\eta, \qquad \rho_e = 2e_h^n - e_h^{n-1} \in V_h, \quad \rho_\eta = 2\eta^n - \eta^{n-1},$$

and $\|\rho_e\|_0^2 \le E^n$. For $i = 1,2,3$ let J_i denote the i-th jump operator in $S_{\mathrm{IMEX}}^{n+1}(w,w)$:

$$J_1 = [\![(\beta\cdot\nabla)w \times n]\!], \qquad J_2 = [\![B_\beta w]\!], \qquad J_3 = [\![\nabla B_\beta w]\!], \qquad B_\beta w = \nabla\times((\beta\cdot\nabla)w).$$

The weight of J_i is h_F^{2i}.

On each element, $(\beta\cdot\nabla)w$, $B_\beta w$ and $\nabla B_\beta w$ are finite sums of terms $M(D^a\beta, D^b w)$. Here M is bilinear with constant coefficients, $a+b=i$ and $b \ge 1$. Explicitly,

$$(a,b) = (0,1) \quad \text{for} \quad i=1, \qquad (a,b) \in \{(0,2),(1,1)\} \quad \text{for} \quad i=2,$$

and

$$(a,b) \in \{(0,3),(1,2),(2,1)\} \quad \text{for} \quad i=3.$$

With the face average $\{\cdot\}$ and

$$[\![M(p,q)]\!] = M([\![p]\!], \{q\}) + M(\{p\}, [\![q]\!])$$

we get

$$|J_i| \le C \sum_{a+b=i,\ b\ge1} \Big(|[\![D^a\beta]\!]|\,|\{D^b w\}| + |\{D^a\beta\}|\,|[\![D^b w]\!]| \Big). \tag{23}$$

The continuity of $\beta \in V_h$ implies $[\![\beta]\!] = 0$. Moreover, $\beta_* \in H^{k+1}$ with $k \geq 2$ gives $D^a\beta_* \in H^1(\Omega)$ for $a \leq 2$, so these derivatives have single-valued traces. Thus, $[\![D^a\beta]\!] = [\![D^a r]\!]$ for $a = 1, 2$. Similarly, for $b \leq \min(k, 3)$, $u \in H^{k+1}$ gives $D^b u^{n+1} \in H^1(\Omega)$, which therefore has single-valued traces. Using $w = u^{n+1} - \nabla \times \xi$, we obtain $[\![D^b w]\!] = -[\![D^b \nabla \times \xi]\!]$. Finally, since w is a polynomial of degree at most k on each element, $D^b w = 0$ for $b > k$, and hence $[\![D^b w]\!] = 0$.

Let $C_u := 3\|u\|_{L^\infty(0,T;W^{2,\infty})}$. Then $|D^a\beta_*| \leq C_u$ for $a \leq 2$. By the inverse inequality, $\|D^b w\|_{0,\infty,K} \leq Ch^{1-b}C_\nabla$ on every element K. The local trace inequality $\|\varphi\|^2_{0,F} \leq C(h^{-1}\|\varphi\|^2_{0,K} + h\|\nabla\varphi\|^2_{0,K})$ and inverse inequalities for ρ_e give, for $a \leq 2$,

$$\sum_F \|D^a\rho_e\|^2_{0,F} \leq Ch^{-2a-1}\|\rho_e\|^2_0 \leq Ch^{-2a-1}E^n, \tag{24}$$

$$\sum_F \|D^a\rho_\eta\|^2_{0,F} \leq C\big(h^{-1}\|D^a\rho_\eta\|^2_{0,h} + h\|D^{a+1}\rho_\eta\|^2_{0,h}\big) \leq Ch^{2k+1-2a}, \tag{25}$$

$$\sum_F \|[\![D^b\nabla \times \xi]\!]\|^2_{0,F} \leq C\big(h^{-1}\|D^{b+1}\xi\|^2_{0,h} + h\|D^{b+2}\xi\|^2_{0,h}\big), \qquad b \leq k. \tag{26}$$

In (25) we used $\|D^j\eta\|_{0,h} \leq \|D^{j+1}\xi\|_{0,h} \leq Ch^{k+1-j}$ for $j \leq 3$, which follows from assumption 4 because $j + 1 \leq 4 \leq \min(5, k + 2)$. In (26), $b + 2 \leq \min(5, k + 2)$.

Consider first the terms in (23) containing r, namely $[\![D^a r]\!]\{D^b w\}$ with $a \geq 1$ and $\{D^a r\}[\![D^b w]\!]$. For these we use $|\{D^b w\}| + |[\![D^b w]\!]| \leq Ch^{1-b}C_\nabla$ together with (24)–(25). Since $a + b = i$, their contribution is bounded by

$$Ch^{2i}\, h^{2-2b}C^2_\nabla\big(h^{-2a-1}E^n + h^{2k+1-2a}\big) = CC^2_\nabla\, h\big(E^n + h^{2k+2}\big).$$

The remaining terms are $D^a\beta_*\,[\![D^b w]\!] = -D^a\beta_*\,[\![D^b\nabla \times \xi]\!]$ with $b \leq \min(i, k)$. By (26) they contribute at most

$$CC^2_u\big(h^{2i-1}\|D^{b+1}\xi\|^2_{0,h} + h^{2i+1}\|D^{b+2}\xi\|^2_{0,h}\big) \leq CC^2_u\|\xi\|^2_*,$$

because $i \geq b$ and $h \leq 1$ give $h^{2i-1} \leq h \cdot h^{2(b+1)-4}$ and $h^{2i+1} \leq h \cdot h^{2(b+2)-4}$, and these are weights occurring in (14). Multiplying by δ_i, summing over i and over the faces, and dividing by $\|\beta\|_{0,\infty} \geq u_{\min}$ proves the first inequality in (22). The second follows from (18) and $h^{2k+3} \leq h^{2k+1}$. □

Remark 1. In (22) the transport field enters only through β_*, which is smooth, and through the discrete remainder r. Consequently the extrapolation defect $\beta_* - u^{n+1} = O(\Delta t^2)$ produces no separate term, and no time-step restriction is needed. The factor h in front of E^n comes from the terms in which r is differentiated or multiplied by a jump of $D^b w_h$, compare [16, Lemma 3.2] for the semi-discrete analogue. The division by $\|\beta\|_{0,\infty}$ is where the lower bound $u_{\min}$ enters, see also [16, Remark 3.4].

3.4 A preliminary estimate and the two-sided bound on the transport field

The next lemma is the fully discrete analogue of the pressure-robust Galerkin estimate of [22] and of [16, Prop. 3.3], with the stabilization included; we give the short proof for completeness.

Lemma 3 (Preliminary robust estimate)**.** *Assume the regularity and approximation hypotheses of Section 3.2, but not* (11)*. Assume also Assumption 1, $k \geq d$, $u_h^0, u_h^1 \in V_h^{\mathrm{div}}$, $\|e_h^0\|_0 + \|e_h^1\|_0 \leq c_1(h^k + \Delta t^2)$, and $\|\tilde{u}_h^{j+1}\|_{0,\infty} \geq u_{\min}$ for $1 \leq j \leq n$. Here*

$$E^m = \|e_h^m\|^2_0 + \|2e_h^m - e_h^{m-1}\|^2_0$$

is the BDF2 energy. Then there are $C_* > 0$ and $\Delta t_1 > 0$, independent of h, Δt, n, $\nu \le \nu_{\max}$ and the pressure, such that for $\Delta t \le \Delta t_1$

$$E^m \le C_*^2 \big(h^k + \Delta t^2\big)^2, \qquad 1 \le m \le n+1.$$

Proof. Fix $1 \le j \le n$ and test (13) (at level $j+1$) with $v_h = e_h^{j+1} \in V_h^{\mathrm{div}}$; the pressure does not appear. We bound the terms as follows:

- the BDF2 term from below by Lemma 1(i);
- the temporal residuals and the viscous projection term by Lemma 1(ii)–(iii);
- the nonlinear residual by Lemma 1(v)–(vi);
- the stabilization residual by

$$|S^{j+1}(e_h^{j+1})| \le \tfrac{1}{2} S^{j+1}_{\mathrm{IMEX}}(e_h^{j+1}, e_h^{j+1}) + \tfrac{1}{2} S^{j+1}_{\mathrm{IMEX}}(w_h^{j+1}, w_h^{j+1}), \tag{27}$$

  where the second term is estimated by Lemma 2, using $\|\tilde{u}_h^{j+1}\|_{0,\infty} \ge u_{\min}$ and $h^{2k+1} \le h^{2k}$.

This yields

$$\begin{aligned}\frac{1}{4\Delta t}(E^{j+1} - E^j) + \frac{3\nu}{4}\|\nabla e_h^{j+1}\|_0^2 + \frac{1}{2} S^{j+1}_{\mathrm{IMEX}}(e_h^{j+1}, e_h^{j+1}) &\le C\,(E^{j+1} + E^j) + C\,(h^{2k} + \Delta t^4) \\ &\quad + \tfrac{1}{2}\big(\|\tau^{j+1}\|_0^2 + \|\rho^{j+1}\|_0^2\big).\end{aligned} \tag{28}$$

Multiplying (28) by $4\Delta t$ gives

$$\begin{aligned}&E^{j+1} - E^j + \Delta t\,\big(3\nu\|\nabla e_h^{j+1}\|_0^2 + 2S^{j+1}_{\mathrm{IMEX}}(e_h^{j+1}, e_h^{j+1})\big) \\ &\qquad \le C\,\Delta t\,(E^{j+1} + E^j) + C\,\Delta t\,(h^{2k} + \Delta t^4) + 2\Delta t\big(\|\tau^{j+1}\|_0^2 + \|\rho^{j+1}\|_0^2\big).\end{aligned} \tag{29}$$

Sum (29) over $j = 1, \dots, m-1$ with $m \le n+1$. Use the residual bounds of the error-splitting subsection,

$$\Delta t \sum_{j=1}^{N-1} \big(\|\tau^{j+1}\|_0^2 + \|\rho^{j+1}\|_0^2\big) \le C(\Delta t^4 + h^{2k+2}),$$

and $E^1 = \|e_h^1\|_0^2 + \|2e_h^1 - e_h^0\|_0^2 \le 9\|e_h^1\|_0^2 + 2\|e_h^0\|_0^2 \le 9c_1^2(h^k + \Delta t^2)^2$. The discrete Gronwall lemma, valid for $C\Delta t \le \frac{1}{2}$, then yields

$$E^m \le C\,(h^{2k} + \Delta t^4) \le C_*^2\,(h^k + \Delta t^2)^2.$$

The constant depends on T, c_1 and the constants of Lemmas 1 and 2. □

Corollary 1 (Two-sided bound on the transport field)**.** *Let $d = 2$, $k \in \{2, 3\}$, $h \le 1$, and $\Delta t \le \min(\Delta t_1, c_0 h^{1/2+\sigma})$ for fixed $c_0 > 0$ and $\sigma \in (0, \frac{1}{2}]$, and let the hypotheses of Lemma 3, except the lower bound on $\tilde{u}_h$, hold. There is $h_0 > 0$, depending on $u_{\min}$, C_*, c_0, σ and norms of u but not on $\nu \le \nu_{\max}$, with the following property. For $h \le h_0$,*

$$u_{\min} \le \|\tilde{u}_h^{n+1}\|_{0,\infty} \le C_\infty := \|u\|_{L^\infty(0,T;L^\infty)} + u_{\min} \qquad (1 \le n \le N-1),$$

and $E^n \le C_*^2(h^k + \Delta t^2)^2$ for $1 \le n \le N$.

Proof. Without loss of generality $C_* \geq 3c_1$. For $1 \leq j \leq N-1$,

$$\tilde{u}_h^{j+1} - u^{j+1} = (2e_h^j - e_h^{j-1}) - (2\eta^j - \eta^{j-1}) + (2u^j - u^{j-1} - u^{j+1}).$$

If $E^j \leq C_*^2(h^k + \Delta t^2)^2$, then an inverse inequality, the pointwise approximation property $\|\eta^j\|_{0,\infty} \leq Ch^k$ (for $d = 2$) and Taylor expansion give

$$\|\tilde{u}_h^{j+1} - u^{j+1}\|_{0,\infty} \leq C_{\mathrm{inv}} h^{-1} C_*(h^k + \Delta t^2) + Ch^k + \Delta t^2 \|\partial_{tt} u\|_{L^\infty(L^\infty)} \leq K(h), \tag{30}$$

with

$$K(h) := C_{\mathrm{inv}} C_* \big(h^{k-1} + c_0^2 h^{2\sigma}\big) + Ch^k + c_0^2 h^{1+2\sigma} \|\partial_{tt} u\|_{L^\infty(L^\infty)}.$$

Since $k \geq 2$ and $\sigma > 0$, $K(h) \to 0$ as $h \to 0$. Choose h_0 with $K(h_0) \leq u_{\min}$. Then (30) and $\|u^{j+1}\|_{0,\infty} \geq 2u_{\min}$ imply

$$u_{\min} \leq \|\tilde{u}_h^{j+1}\|_{0,\infty} \leq C_\infty.$$

We now argue by induction on n that the two-sided bound holds for $1 \leq j \leq n$. For $n = 1$,

$$E^1 \leq 9c_1^2(h^k + \Delta t^2)^2 \leq C_*^2(h^k + \Delta t^2)^2,$$

and the preceding argument with $j = 1$ applies. If the lower bound holds for $1 \leq j \leq n$, Lemma 3 gives

$$E^{n+1} \leq C_*^2(h^k + \Delta t^2)^2,$$

with C_* independent of n, and the preceding argument with $j = n+1$ gives the two-sided bound for $\tilde{u}_h^{n+2}$. □

Remark 2 (On the time-step condition). The condition $\Delta t \leq c_0 h^{1/2+\sigma}$ enters only in Corollary 1, where it makes $K(h)$ small: for $d = 2$ it gives $h^{-1}\Delta t^2 \leq c_0^2 h^{2\sigma} \to 0$, and nothing more is needed. It is weaker than the more common $\Delta t \leq c_0 h$. Theorem 2 uses the corollary only through the bounds $u_{\min} \leq \|\tilde{u}_h^{n+1}\|_{0,\infty} \leq C_\infty$. If the normalization of the stabilization is clamped, these bounds are not needed for the velocity, and the condition disappears from the velocity estimate (Remark 4). In our experiments (Section 5) we did not observe any time-step restriction.

Lemma 4 (Nonlinear residual estimate). *Let $d = 2$ and $k \in \{2, 3\}$. Assume the regularity, projection, exact-divergence and recovery hypotheses of Section 3.2, and the upper bound $\|\tilde{u}_h^{n+1}\|_{0,\infty} \leq C_\infty$ of Corollary 1. For every $\varepsilon > 0$,*

$$\begin{aligned} |N^{n+1}(e_h^{n+1})| \leq \varepsilon S_{\mathrm{IMEX}}^{n+1}(e_h^{n+1}, e_h^{n+1}) + C\big(E^{n+1} + E^n\big) \\ + C\big(\Delta t^4 + h^{2k+2}\big) + C_\varepsilon \|z^{n+1} - P_h z^{n+1}\|_*^2, \end{aligned} \tag{31}$$

where $E^j = \|e_h^j\|_0^2 + \|2e_h^j - e_h^{j-1}\|_0^2$. The constant C_ε depends on ε^{-1}, $\delta_{\min}^{-1} := (\min_j \delta_j)^{-1}$ and C_∞.

Proof. By Lemma 1(v), it suffices to bound the convection-dominated term $((\tilde{u}_h^{n+1} \cdot \nabla)\eta^{n+1}, e_h^{n+1})$ by the right-hand side of (31). We next detail the last, convection-dominated term. Set, only in this derivation,

$$\beta = \tilde{u}_h^{n+1}, \qquad e = e_h^{n+1}, \qquad \xi = z^{n+1} - P_h z^{n+1}, \qquad \eta^{n+1} = \nabla \times \xi,$$

and let $B_\beta e = \nabla \times ((\beta \cdot \nabla) e)$ elementwise. Because $\nabla \cdot \beta = 0$ and $\beta = 0$ on $\partial\Omega$, integration by parts first in the transport term and then elementwise in the curl identity yields

$$\begin{aligned} ((\beta \cdot \nabla)\eta^{n+1}, e) &= -((\beta \cdot \nabla)e, \nabla \times \xi)_h \\ &= -\sum_{F \in \mathcal{F}_h^i} \langle [\![(\beta \cdot \nabla)e \times n]\!], \xi \rangle_F - (B_\beta e, \xi)_h. \end{aligned} \tag{32}$$

Only absolute values will be used, so the signs in (32) are immaterial. Denote the face and volume terms on its right by T_F and T_B, respectively.

For T_F, weighted Cauchy–Schwarz, the definition of S_1, and the local trace inequality give

$$|T_F| \le \|h[\![(\beta \cdot \nabla)e \times n]\!]\|_{0,\mathcal{F}_h^i} \|h^{-1}\xi\|_{0,\mathcal{F}_h^i}, \tag{33}$$

$$\|h^{-1}\xi\|_{0,\mathcal{F}_h^i}^2 \le C\left(h^{-3}\|\xi\|_{0,h}^2 + h^{-1}\|\nabla\xi\|_{0,h}^2\right). \tag{34}$$

By assumption, $\|\beta\|_{0,\infty} \le C_\infty$. Since S_{IMEX}^{n+1} is the raw jump form divided by $\|\beta\|_{0,\infty}$, we have

$$\|h[\![(\beta \cdot \nabla)e \times n]\!]\|_{0,\mathcal{F}_h^i}^2 \le \delta_1^{-1}\|\beta\|_{0,\infty} S_{\mathrm{IMEX}}^{n+1}(e,e).$$

Young's inequality in (33), followed by (34) and $h^{-3}\|\xi\|_{0,h}^2 + h^{-1}\|\nabla\xi\|_{0,h}^2 \le \|\xi\|_*^2$, gives

$$|T_F| \le \varepsilon S_{\mathrm{IMEX}}^{n+1}(e,e) + \frac{C\,C_\infty}{\varepsilon\,\delta_1}\left(h^{-3}\|\xi\|_{0,h}^2 + h^{-1}\|\nabla\xi\|_{0,h}^2\right) \le \varepsilon S_{\mathrm{IMEX}}^{n+1}(e,e) + C_\varepsilon\|\xi\|_*^2, \tag{35}$$

with $C_\varepsilon = CC_\infty/(\varepsilon\delta_1)$.

It remains to control $T_B = (B_\beta e, \xi)_h$. Since $k \in \{2,3\}$, $B_\beta e$ is a broken polynomial of degree $2k-2 \le k+1$, so the recovery property (11) applies to $\chi_h = B_\beta e$ (this is the case $C_{u,h} = 0$ in [16, Lemma 3.6]). The L^2-orthogonality of P_h implies $(m_h, \xi) = 0$ for every $m_h \in Z_h$. Therefore

$$|(B_\beta e, \xi)_h| = |(B_\beta e - m_h, \xi)_h| \le \|h^{3/2}(B_\beta e - m_h)\|_{0,h}\, \|h^{-3/2}\xi\|_{0,h}.$$

Take the infimum over $m_h \in Z_h$ and apply (11). The definitions of S_2 and S_3 yield

$$\inf_{m_h \in Z_h} \|h^{3/2}(B_\beta e - m_h)\|_{0,h} \le C(C_\infty/\delta_{\min})^{1/2} S_{\mathrm{IMEX}}^{n+1}(e,e)^{1/2}. \tag{36}$$

Since $\|h^{-3/2}\xi\|_{0,h}^2 \le \|\xi\|_*^2$, Young's inequality gives

$$|T_B| \le \varepsilon S_{\mathrm{IMEX}}^{n+1}(e,e) + C_\varepsilon\|\xi\|_*^2. \tag{37}$$

Combining (32), (35), and (37), and replacing 2ε by ε, proves the required estimate

$$|((\tilde{u}_h^{n+1} \cdot \nabla)\eta^{n+1}, e_h^{n+1})| \le \varepsilon S_{\mathrm{IMEX}}^{n+1}(e_h^{n+1}, e_h^{n+1}) + C_\varepsilon\|z^{n+1} - P_h z^{n+1}\|_*^2 + C\|e_h^{n+1}\|_0^2. \tag{38}$$

Combining (38) with Lemma 1(v) and $\|e_h^{n+1}\|_0^2 \le E^{n+1}$ proves (31). □

3.5 Fully discrete error estimate

Theorem 2 (Fully discrete velocity error estimate)**.** *Let (u,p) be a sufficiently smooth solution of (1), and let (u_h^n, p_h^n) be the BDF2–IMEX approximation (8). Assume the structural finite element*

hypotheses of Section 3.2, $d = 2$, $k \in \{2, 3\}$, Assumption 1, the stated temporal regularity, and $u_h^0, u_h^1 \in V_h^{\mathrm{div}}$. If the starting values satisfy

$$\|u(t_j) - u_h^j\|_0 \le C(\varepsilon_h^{1/2} + \Delta t^2), \qquad j = 0, 1,$$

then, for $1 \le m \le N$,

$$\|u(t_m) - u_h^m\|_0^2 + \nu\Delta t \sum_{n=1}^{m-1} \|\nabla e_h^{n+1}\|_0^2 + \Delta t \sum_{n=1}^{m-1} S_{\mathrm{IMEX}}^{n+1}(e_h^{n+1}, e_h^{n+1}) \le C\big(\varepsilon_h + \Delta t^4\big),$$

provided h and Δt are sufficiently small and $\Delta t \le c_0 h^{1/2+\sigma}$ for a fixed $\sigma \in (0, \frac{1}{2}]$. Here $\varepsilon_h = (\nu + h)h^{2k}$. The constant C may depend on norms of the exact velocity, the final time, the shape-regularity of the mesh, and the stabilization parameters, but it is independent of the pressure and of ν^{-1}. For $d = 3$ and $k = 3$ the same estimate holds, with $\Delta t \le c_0 h^{3/4+\sigma}$, provided (11) and assumption 4 of Section 3.2 hold for the potential spaces of [13] and their L^2-projection.

Proof. By Corollary 1, $u_{\min} \le \|\tilde{u}_h^{n+1}\|_{0,\infty} \le C_\infty$ holds at every level. Hence Lemmas 1, 2 and 4 are available at every step.

Recall the splitting $u(t_n) - u_h^n = \eta^n - e_h^n$, with $e_h^n \in V_h^{\mathrm{div}}$. Take $v_h = e_h^{n+1}$ in (13). There is no pressure term in that identity: the continuous pressure gives zero against a divergence-free test function, and the discrete pressure satisfies $(p_h^{n+1}, \nabla \cdot e_h^{n+1}) = 0$. This is the pressure-robust step of the argument.

By Lemma 4 and (18), for every $\varepsilon > 0$,

$$|N^{n+1}(e_h^{n+1})| \le \varepsilon S_{\mathrm{IMEX}}^{n+1}(e_h^{n+1}, e_h^{n+1}) + C(E^{n+1} + E^n) + C_\varepsilon(\varepsilon_h + \Delta t^4). \tag{39}$$

For the stabilization residual, Cauchy's inequality in the stabilization seminorm and Lemma 2 give

$$\begin{aligned} |S^{n+1}(e_h^{n+1})| &\le \varepsilon S_{\mathrm{IMEX}}^{n+1}(e_h^{n+1}, e_h^{n+1}) + C_\varepsilon S_{\mathrm{IMEX}}^{n+1}(w_h^{n+1}, w_h^{n+1}) \\ &\le \varepsilon S_{\mathrm{IMEX}}^{n+1}(e_h^{n+1}, e_h^{n+1}) + C_\varepsilon\big(hE^n + h^{2k+1}\big) \le \varepsilon S_{\mathrm{IMEX}}^{n+1}(e_h^{n+1}, e_h^{n+1}) + C_\varepsilon\big(hE^n + \varepsilon_h\big). \end{aligned} \tag{40}$$

Insert Lemma 1(i)–(iii), (39) and (40) into (13) with $v_h = e_h^{n+1}$. Choose $\varepsilon = \frac{1}{4}$, so that the two stabilization contributions on the right are absorbed by the left-hand side. Then multiply by $4\Delta t$. We obtain

$$\begin{aligned} E^{n+1} - E^n + \kappa\Delta t\big(\nu\|\nabla e_h^{n+1}\|_0^2 + S_{\mathrm{IMEX}}^{n+1}(e_h^{n+1}, e_h^{n+1})\big) \\ \le C\Delta t\big(E^{n+1} + E^n\big) + C\Delta t\big(\varepsilon_h + \Delta t^4\big) + 2\Delta t\big(\|\tau^{n+1}\|_0^2 + \|\rho^{n+1}\|_0^2\big), \end{aligned} \tag{41}$$

with a constant $\kappa > 0$ independent of h, Δt, ν and the pressure. Sum (41) from $n = 1$ to $m - 1$. The BDF2 energies telescope, giving

$$\begin{aligned} E^m + \kappa\Delta t \sum_{n=1}^{m-1} \big(\nu\|\nabla e_h^{n+1}\|_0^2 + S_{\mathrm{IMEX}}^{n+1}(e_h^{n+1}, e_h^{n+1})\big) \\ \le E^1 + C\Delta t \sum_{n=1}^{m-1} (E^{n+1} + E^n) + C(\varepsilon_h + \Delta t^4) + 2\Delta t \sum_{n=1}^{m-1} \big(\|\tau^{n+1}\|_0^2 + \|\rho^{n+1}\|_0^2\big). \end{aligned} \tag{42}$$

The residual estimates of the error-splitting subsection give

$$\Delta t \sum_{n=1}^{m-1} \big(\|\tau^{n+1}\|_0^2 + \|\rho^{n+1}\|_0^2\big) \le C(\Delta t^4 + h^{2k+2}) \le C(\Delta t^4 + \varepsilon_h).$$

The assumed accuracy of the starting values gives $E^1 \le C(\varepsilon_h + \Delta t^4)$. For Δt sufficiently small, the term involving E^{n+1} in the sum is handled by the discrete Gronwall lemma. The Gronwall factor depends on T and on the velocity-dependent constants, but the argument introduces no factor ν^{-1}. Thus

$$E^m + \Delta t \sum_{n=1}^{m-1} \left(\nu \|\nabla e_h^{n+1}\|_0^2 + S_{\mathrm{IMEX}}^{n+1}(e_h^{n+1}, e_h^{n+1}) \right) \le C(\varepsilon_h + \Delta t^4). \tag{43}$$

Finally,

$$\|u(t_m) - u_h^m\|_0 \le \|\eta^m\|_0 + \|e_h^m\|_0, \quad \text{with} \quad \|e_h^m\|_0^2 \le E^m \quad \text{and} \quad \|\eta^m\|_0^2 \le C h^{2k+2} \le C\varepsilon_h.$$

This proves the theorem. □

Remark 3 (On Theorem 2). (i) *Rates.* For $\nu \lesssim h$ we have $\varepsilon_h \simeq h^{2k+1}$, so the bound is of order $h^{k+1/2} + \Delta t^2$, the best L^2 rate known for ν-independent estimates [17]. For fixed $\nu \gg h$ the bound reduces to $\nu^{1/2} h^k + \Delta t^2$, which is worse than the Galerkin-optimal h^{k+1}. The loss comes from a single term. By Lemma 2 and (18), every contribution except the viscous projection term $\nu(\nabla\eta, \nabla e_h)$ is $O(h^{2k+1})$; the factor νh^{2k} is the price we pay for the non-Ritz projection w_h, which in turn is what keeps the constants independent of ν. The same factor $(\nu + h)h^{2k}$ appears in the robust analyses [3, (3.13)], [16, Cor. 3.8], [5] and [6]. For fixed ν, a standard non-robust argument based on the modified Stokes projection s_h removes the term νh^{2k} at the expense of constants that depend on ν^{-1}. Since the consistency error of the stabilization is $O(h^{2k+1})$, we expect this argument to give $C(\nu^{-1})h^{k+1/2}$, and hence an error bound of the form $\min\{(\nu + h)^{1/2} h^k, C(\nu^{-1})h^{k+1/2}\} + \Delta t^2$.

Min-type bound: non-robust branch not written out; check before keeping.

(ii) *Regularity.* We assume that the norms of u and z in Assumption 1 and Section 3.2 are bounded uniformly in ν. As in every ν-robust analysis of this kind, this rules out boundary layers.

(iii) *Starting values.* With $u_h^0 = s_h(0)$ one has $\|u(0) - u_h^0\|_0 \le C h^{k+1}$. For u_h^1 computed by one IMEX-Euler step, the energy argument of the proof applied to a single step gives $\|e_h^1\|_0 \le \|e_h^0\|_0 + C(\Delta t^2 + (\Delta t\, \varepsilon_h)^{1/2})$. In this argument (15) is replaced by $(a-b, a) \ge \frac{1}{2}(\|a\|_0^2 - \|b\|_0^2)$, and the $O(\Delta t)$ truncation error of the Euler step enters multiplied by Δt. The hypothesis on the starting values is therefore satisfied.

(iv) *Dimension.* The recovery property (11) is proved in [3, Lemma 4.1] for $d = 2$ only; for $d = 3$ the theorem is conditional on it.

Remark 4 (A clamped normalization). Suppose the factor $\|\tilde{u}_h^{n+1}\|_{0,\infty}^{-1}$ in S_{IMEX}^{n+1} is replaced by N_{n+1}^{-1}, where

$$N_{n+1} := \min\left\{ \max\{\|\tilde{u}_h^{n+1}\|_{0,\infty}, u_{\min}\}, U_{\max} \right\}, \qquad 0 < u_{\min} < U_{\max},$$

and one chooses $U_{\max} \ge \|u\|_{L^\infty(0,T;L^\infty)}$. The stabilization stays nonnegative, so Theorem 1 still holds. Lemma 2 uses only $N_{n+1} \ge u_{\min}$. In Lemma 4, the upper bound $\|\tilde{u}_h^{n+1}\|_{0,\infty} \le C_\infty$ is needed only to get $\|h[\![(\beta \cdot \nabla)e \times n]\!]\|_{0,\mathcal{F}_h^i}^2 \le \delta_1^{-1} N_{n+1} S_{\mathrm{IMEX}}^{n+1}(e,e)$ and the analogous bound in (36), and with clamping both follow from $N_{n+1} \le U_{\max}$. For $d = 2$ and $k \in \{2, 3\}$, the velocity estimate of Theorem 2 therefore holds for the clamped scheme without Lemma 3 and Corollary 1, that is, without any coupling of Δt and h beyond the Gronwall condition $\Delta t \le \Delta t_0$. The pressure analysis in Section 4 is different: it uses $\|\tilde{u}_h^{n+1}\|_{0,\infty} \le C_\infty$ in R_{conv} and in (7), so it still relies on Corollary 1. Asymptotically, $N_{n+1} \in [u_{\min}, U_{\max}]$ and the two schemes coincide.

Clamped variant: check every use of C_∞ and $u_{\min}$ in Lemmas 3, 5 and Thm 2.

4 Pressure error estimate

Although the velocity estimates above are pressure robust, the pressure itself is recovered through the discrete inf-sup condition. Let $\Pi_h^Q : L_0^2(\Omega) \to Q_h$ denote a pressure projection satisfying the approximation estimate $\|p(t_n) - \Pi_h^Q p(t_n)\|_0 \le Ch^k \|p(t_n)\|_{H^k}$. We split the pressure error as

$$p(t_n) - p_h^n = \zeta_p^n + \xi_p^n, \qquad \zeta_p^n = p(t_n) - \Pi_h^Q p(t_n), \quad \xi_p^n = \Pi_h^Q p(t_n) - p_h^n.$$

The term ζ_p^n is the pressure approximation error, while $\xi_p^n \in Q_h$ is controlled by the momentum residual.

Theorem 3 (Fully discrete pressure estimate)**.** *Assume the hypotheses of Theorem 2 and $p \in L^\infty(0,T;H^k(\Omega))$. Then the following holds.*

(i) (Pointwise in time.) For each $1 \le n \le N-1$,

$$\|p(t_{n+1}) - p_h^{n+1}\|_0 \le Ch^k \|p(t_{n+1})\|_{H^k} + C\|D_t^{\mathrm{BDF2}} e_u^{n+1}\|_{-1,h} + C\nu\|\nabla e_u^{n+1}\|_0 + C\|R_{\mathrm{conv}}^{n+1}\|_{-1,h} + C\|R_{\mathrm{stab}}^{n+1}\|_{-1,h} + C\|\tau^{n+1}\|_{H^{-1}},$$

where $e_u^{n+1} = u(t_{n+1}) - u_h^{n+1}$, and

$$\|D_t^{\mathrm{BDF2}} e_u^{n+1}\|_{-1,h} = \sup_{0 \ne v_h \in V_h} \frac{(D_t^{\mathrm{BDF2}} e_u^{n+1}, v_h)}{\|\nabla v_h\|_0},$$

$$\|R_{\mathrm{conv}}^{n+1}\|_{-1,h} = \sup_{0 \ne v_h \in V_h} \frac{|(u^{n+1}\cdot\nabla u^{n+1}, v_h) - ((\tilde{u}_h^{n+1}\cdot\nabla)u_h^{n+1}, v_h)|}{\|\nabla v_h\|_0},$$

$$\|R_{\mathrm{stab}}^{n+1}\|_{-1,h} = \sup_{0 \ne v_h \in V_h} \frac{|S_{\mathrm{IMEX}}^{n+1}(u_h^{n+1}, v_h)|}{\|\nabla v_h\|_0}.$$

(ii) (Cumulative.)

$$\Delta t \sum_{n=1}^{N-1} \|p(t_{n+1}) - p_h^{n+1}\|_0^2 \le C\big(h^{2k} + \Delta t^4\big) + C\Delta t \sum_{n=1}^{N-1} \|D_t^{\mathrm{BDF2}} e_u^{n+1}\|_{-1,h}^2.$$

(iii) (Robust, suboptimal rate.)

$$\Big(\Delta t \sum_{n=1}^{N-1} \|p(t_{n+1}) - p_h^{n+1}\|_0^2\Big)^{1/2} \le C\Big(h^{k-1} + \frac{\Delta t^2}{h}\Big).$$

Proof. Fix a time level t_{n+1}. The triangle inequality gives

$$\|p^{n+1} - p_h^{n+1}\|_0 \le \|p^{n+1} - \Pi_h^Q p^{n+1}\|_0 + \|\xi_p^{n+1}\|_0.$$

The first term is bounded by the approximation property of Π_h^Q. It remains to estimate ξ_p^{n+1}. By the discrete inf-sup condition,

$$\beta\|\xi_p^{n+1}\|_0 \le \sup_{0 \ne v_h \in V_h} \frac{(\xi_p^{n+1}, \nabla\cdot v_h)}{\|\nabla v_h\|_0}.$$

Since $\xi_p^{n+1} = \Pi_h^Q p^{n+1} - p_h^{n+1}$, we write

$$(\xi_p^{n+1}, \nabla \cdot v_h) = (p^{n+1} - p_h^{n+1}, \nabla \cdot v_h) - (p^{n+1} - \Pi_h^Q p^{n+1}, \nabla \cdot v_h).$$

The second term is bounded by $C\|p^{n+1} - \Pi_h^Q p^{n+1}\|_0 \|\nabla v_h\|_0$. For the first term, subtract the momentum equation in (8) from the exact momentum equation in (2) at t_{n+1}, tested with an arbitrary $v_h \in V_h$. This gives

$$\begin{aligned}(p^{n+1} - p_h^{n+1}, \nabla \cdot v_h) = (\partial_t u^{n+1} - D_t^{\mathrm{BDF2}} u_h^{n+1}, v_h) + \nu(\nabla e_u^{n+1}, \nabla v_h) \\ + (u^{n+1} \cdot \nabla u^{n+1}, v_h) - ((\tilde{u}_h^{n+1} \cdot \nabla) u_h^{n+1}, v_h) - S_{\mathrm{IMEX}}^{n+1}(u_h^{n+1}, v_h).\end{aligned}$$

Now add and subtract $D_t^{\mathrm{BDF2}} u^{n+1}$: $\partial_t u^{n+1} - D_t^{\mathrm{BDF2}} u_h^{n+1} = -\tau^{n+1} + D_t^{\mathrm{BDF2}} e_u^{n+1}$. Dividing by $\|\nabla v_h\|_0$, taking the supremum over $v_h \in V_h$, and using the definitions of the discrete negative norms yields

$$\begin{aligned}\|\xi_p^{n+1}\|_0 \le C\|p^{n+1} - \Pi_h^Q p^{n+1}\|_0 + C\|D_t^{\mathrm{BDF2}} e_u^{n+1}\|_{-1,h} + C\|\tau^{n+1}\|_{H^{-1}} \\ + C\nu\|\nabla e_u^{n+1}\|_0 + C\|R_{\mathrm{conv}}^{n+1}\|_{-1,h} + C\|R_{\mathrm{stab}}^{n+1}\|_{-1,h}.\end{aligned}$$

Combining this with the triangle inequality proves the pointwise-in-time estimate.

It remains to bound the cumulative estimate. Using $(a_1 + \cdots + a_m)^2 \le m(a_1^2 + \cdots + a_m^2)$, multiplying by Δt, and summing from $n = 1$ to $N - 1$ gives a discrete $L^2(0, T; L^2(\Omega))$-in-time pressure bound. The pressure projection term is estimated by

$$\Delta t \sum h^{2k} \|p^{n+1}\|_{H^k}^2 \le C h^{2k} \|p\|_{L^\infty(0,T;H^k)}^2.$$

The BDF2 truncation residual satisfies $\Delta t \sum \|\tau^{n+1}\|_{H^{-1}}^2 \le C\Delta t^4$, by second-order consistency. The accumulated velocity-gradient contribution is controlled by splitting the total error as $e_u^{n+1} = \eta^{n+1} - e_h^{n+1}$. Indeed, since $\nu \le \nu_{\max}$,

$$\Delta t \sum_{n=1}^{N-1} \nu^2 \|\nabla e_u^{n+1}\|_0^2 \le C\Delta t \sum_{n=1}^{N-1} \left(\nu^2 \|\nabla e_h^{n+1}\|_0^2 + \nu^2 \|\nabla \eta^{n+1}\|_0^2\right) \le C(h^{2k} + \Delta t^4),$$

where the first term is bounded by the discrete velocity estimate and the second by $\|\nabla \eta^{n+1}\|_0 \le Ch^k$.

The convective residual is split as

$$(u^{n+1} \cdot \nabla u^{n+1} - \tilde{u}_h^{n+1} \cdot \nabla u_h^{n+1}, v_h) = ((u^{n+1} - \tilde{u}_h^{n+1}) \cdot \nabla u^{n+1}, v_h) + (\tilde{u}_h^{n+1} \cdot \nabla e_u^{n+1}, v_h).$$

The first term is bounded as

$$((u^{n+1} - \tilde{u}_h^{n+1}) \cdot \nabla u^{n+1}, v_h) \le \|u^{n+1} - \tilde{u}_h^{n+1}\|_0 \|\nabla u^{n+1}\|_{0,\infty} \|v_h\|_0,$$

with $u^{n+1} - \tilde{u}_h^{n+1} = (u^{n+1} - 2u^n + u^{n-1}) + 2e_u^n - e_u^{n-1}$, followed by the Poincaré inequality $\|v_h\|_0 \le C_P \|\nabla v_h\|_0$. For the second term the gradient must be moved onto v_h: since $\nabla \cdot \tilde{u}_h^{n+1} = 0$,

$$(\tilde{u}_h^{n+1} \cdot \nabla e_u^{n+1}, v_h) = -(e_u^{n+1}, (\tilde{u}_h^{n+1} \cdot \nabla) v_h) \le \|\tilde{u}_h^{n+1}\|_{0,\infty} \|e_u^{n+1}\|_0 \|\nabla v_h\|_0,$$

and $\|\tilde{u}_h^{n+1}\|_{0,\infty} \le C_\infty$ by Corollary 1. (Without this integration by parts the term contains ∇e_u and is not bounded by $\|e_u\|_0$.) Hence

$$\|R_{\mathrm{conv}}^{n+1}\|_{-1,h} \le C\Delta t^2 + C\big(\|e_u^{n+1}\|_0 + \|e_u^n\|_0 + \|e_u^{n-1}\|_0\big),$$

and, since $\|u^j - u_h^j\|_0^2 \le C(\varepsilon_h + \Delta t^4)$,

$$\Delta t \sum \|R_{\mathrm{conv}}^{n+1}\|_{-1,h}^2 \le C(\varepsilon_h + \Delta t^4) \le C(h^{2k} + \Delta t^4).$$

A pointwise-in-time bound is not available: the velocity estimate controls $\Delta t \sum S_{\mathrm{IMEX}}(e_h, e_h)$, not $S_{\mathrm{IMEX}}(e_h^{n+1}, e_h^{n+1})$ at a fixed level. Here the normalized continuity estimate (7) is used. Cauchy–Schwarz in the stabilization seminorm gives

$$|S_{\mathrm{IMEX}}^{n+1}(u_h^{n+1}, v_h)| \le S_{\mathrm{IMEX}}^{n+1}(u_h^{n+1}, u_h^{n+1})^{1/2} S_{\mathrm{IMEX}}^{n+1}(v_h, v_h)^{1/2},$$

and (7) therefore yields

$$\|R_{\mathrm{stab}}^{n+1}\|_{-1,h} \le C\big(h\|\tilde{u}_h^{n+1}\|_{0,\infty}\big)^{1/2} S_{\mathrm{IMEX}}^{n+1}(u_h^{n+1}, u_h^{n+1})^{1/2}.$$

Moreover,

$$S_{\mathrm{IMEX}}(u_h, u_h) \le 2S_{\mathrm{IMEX}}(e_h, e_h) + 2S_{\mathrm{IMEX}}(w_h, w_h).$$

Summing, and using (43) for the first term and Lemma 2 together with (43) for the second, gives

$$\Delta t \sum \|R_{\mathrm{stab}}^{n+1}\|_{-1,h}^2 \le Ch(\varepsilon_h + \Delta t^4) \le C(h^{2k} + \Delta t^4).$$

Thus all terms are of order $h^{2k} + \Delta t^4$, except for the discrete negative-norm time derivative of the velocity error, which is kept explicitly. This gives

$$\Delta t \sum \|p^{n+1} - p_h^{n+1}\|_0^2 \le C(h^{2k} + \Delta t^4) + C\Delta t \sum \|D_t^{\mathrm{BDF2}} e_u^{n+1}\|_{-1,h}^2.$$

The suboptimal bound then follows from Proposition 1 in Appendix A. □

Remark 5 (Towards an optimal pressure rate). Theorem 3(iii) loses one power of h, and the only reason is the inverse inequality in Proposition 1. In practice the loss is limited: under the step-size condition $\Delta t \le c_0 h^{1/2+\sigma}$ of Theorem 2 the temporal part satisfies $\Delta t^2/h \le c_0^2 h^{2\sigma}$, and for the scaling $\Delta t \propto h^{3/2}$ used in Section 5 the bound (iii) is of order h for $k = 2$ and h^2 for $k = 3$. If we knew that

$$\Delta t \sum_{n=1}^{N-1} \|D_t^{\mathrm{BDF2}} e_h^{n+1}\|_{-1,h}^2 \le C\big(h^{2k} + \Delta t^4\big),$$

then Theorem 3(ii) would give the optimal rate $h^k + \Delta t^2$, since $\|D_t^{\mathrm{BDF2}} e_u^{n+1}\|_{-1,h} \le C\|\rho^{n+1}\|_0 + \|D_t^{\mathrm{BDF2}} e_h^{n+1}\|_{-1,h}$ and $\Delta t \sum \|\rho^{n+1}\|_0^2 \le Ch^{2k+2}$. Proving such a bound requires an H^1-level estimate. One would test the error equation with $D_t^{\mathrm{BDF2}} e_h^{n+1} \in V_h^{\mathrm{div}}$, use a G-stability bound for first differences of BDF2 as in [14], and control the consistency residuals of the convective and CIP terms in L^2 without losing ν-robustness. For the semi-discrete CIP scheme, [16, Remark 3.9] points to a related route through a negative-norm bound on $\partial_t e_h$, following [12]. We leave this question for future work.

5 Numerical studies

We now consider three test problems. Example 1 (Section 5.1) is a Taylor–Green vortex with manufactured forcing; we use it for the spatial and temporal convergence, the robustness with respect to ν and to the pressure, the role of the three CIP terms and the time-step restriction.

Section 5.2 shows a three-dimensional travelling vortex computed with a variant of the scheme, and Section 5.3 a deliberately under-resolved shear layer, which shows what the stabilization does to spurious oscillations.

The experiments use the Scott–Vogelius (SV) pair $P_k/P_{k-1}^{\mathrm{disc}}$ on Alfeld-split meshes, which gives pointwise divergence-free velocities, with the Galerkin and the CIP-stabilized method. Where a comparison is informative — the comparison of the discretizations, the ν-sweep, the pressure-robustness tests, and Sections 5.2 and 5.3 — we include, as a reference, the inf-sup stable Taylor–Hood (TH) pair P_k/P_{k-1} on standard meshes, which satisfies the divergence constraint only weakly; where stated, it is combined with grad-div stabilization with $\gamma = 0.1$ [12]. For SV, grad-div would have no effect, since the discrete velocity is already divergence-free.

Meshes consist of n^d squares or cubes of size $h = 1/n$. Exact and discrete pressures are normalized to zero mean before pressure errors are computed. The time step is given with each example; in Example 1 it is fixed, $\Delta t = 5 \cdot 10^{-3}$, small enough that the temporal error stays below the spatial one on all meshes shown. [Authors: redraw Figures 4–8 in the style of the new figures (Computer Modern, ticks and legends 8 pt, labels 9 pt): `print_figures.py` in galerkit does this from the saved results.]

To study convergence in time, we fix the mesh and vary the number of steps N on $[0, T]$, so that $\Delta t = T/N$. The temporal error is measured against a reference solution on the same mesh with four times the largest N; a second-order method should then show slope -2 when the error is plotted against N.

Pressure robustness is tested as in [21]: the size of the pressure is varied while the velocity is kept fixed, so a pressure-robust method should produce the same discrete velocity for every pressure. In Example 1 and in the three-dimensional vortex we add $R\nabla\phi$ to the body force, with $\phi = (2\pi)^{-1} \prod_{i=1}^{d} \sin(2\pi x_i)$, which changes the pressure from p to $p + R\phi$. The gradient part of the force is assembled as $-(R\phi, \nabla \cdot v_h)$, which equals $(R\nabla\phi, v_h)$ for exact integration and vanishes for every divergence-free test function. If the gradient is integrated by quadrature instead, the SV velocity moves by the quadrature error, about $10^{-5}R$ on a coarse mesh; this is not a property of the scheme.

In all computations the stabilization is normalized by $\max(u_{\min}, \|\tilde{u}_h^{n+1}\|_{0,\infty})^{-1}$ with $u_{\min} = 10^{-2}$, the variant discussed after (7), and $\|\tilde{u}_h^{n+1}\|_{0,\infty}$ is evaluated as the maximum over the Lagrange nodes. Experiment (E5) below compares this with the original and the clamped normalization.

5.1 Example 1: Taylor–Green vortex

On $\Omega = (0,1)^2$ we consider the generalized Taylor–Green vortex

$$\psi(x,y,t) = \pi^{-1} \sin(\pi x) \sin(2\pi y)\, e^{-5\pi^2 \nu t}, \qquad u = \nabla \times \psi = e^{-5\pi^2 \nu t} \begin{pmatrix} 2\sin(\pi x)\cos(2\pi y) \\ -\cos(\pi x)\sin(2\pi y) \end{pmatrix},$$

and compute f from (1). The field satisfies $\nabla \cdot u = 0$ and $u \cdot n = 0$ on $\partial\Omega$, but its tangential velocity does not vanish there: the computations impose the Dirichlet data of the exact velocity, so this example lies outside the no-slip setting of Theorems 2 and 3 and probes the scheme beyond the letter of the analysis. We have $\|u(t)\|_{0,\infty} = 2e^{-5\pi^2\nu t}$, bounded away from zero on $[0, T]$ (Assumption 1), and for small ν the solution is quasi-steady, with the forcing balancing the genuinely nonlinear convective term. We use $T = 0.5$, $\nu = 10^{-6}$ unless stated otherwise, SV on the Alfeld split of a crisscross mesh, TH on the crisscross mesh without the split, CIP parameters $(\delta_1, \delta_2, \delta_3) = (10^{-1}, 10^{-2}, 10^{-3})$, grad-div parameter $\gamma = 0.1$, and u_h^0 the modified Stokes projection of $u(0)$. The time step is $\Delta t = 5 \cdot 10^{-3}$ for every mesh.

Taylor–Green setup: confirm the orientation of ψ (x against y), the pressure used by the code, and the boundary condition as implemented.

CIP weights: every Taylor–Green study records $(\delta_1, \delta_2, \delta_3) = s\,(10^{-1}, 10^{-2}, 10^{-3})$ with $\delta_3 = \delta_1$ in the reweighted variant of (E4), while the shear-layer runs of Section 5.3 use $s\,(10^{-2}, 10^{-4}, 10^{-6})$

Without stabilization. Table 1 shows the two Galerkin discretizations, SV and TH without any stabilization, at $\nu = 10^{-6}$. The results are poor: both pairs converge only with orders near 2 in L^2 and near 1 in H^1, below the $h^{k+1/2}$ of the stabilized method, their errors at $n = 64$ lie one to two orders of magnitude above the stabilized values below, and TH in addition carries an $O(1)$ divergence error. The remaining experiments quantify what the stabilization adds.

Table 1: Example 1, $k = 2$, $\nu = 10^{-6}$: the unstabilized (Galerkin) methods on all meshes (numbers in brackets: orders between consecutive meshes).

Method	n	$\|u - u_h\|_0$	$\|u - u_h\|_1$	$\|p - p_h\|_0$	$\|\nabla \cdot u_h\|_0$
SV, Galerkin	8	2.25e-2	3.26	1.32e-2	2e-14
SV, Galerkin	16	6.07e-3 (1.89)	1.74 (0.90)	3.95e-3 (1.74)	4e-14
SV, Galerkin	32	1.53e-3 (1.99)	0.789 (1.14)	9.45e-4 (2.06)	7e-14
SV, Galerkin	64	3.44e-4 (2.16)	0.298 (1.41)	2.04e-4 (2.21)	2e-13
TH, Galerkin	8	1.47e-1	11.5	4.81e-2	7.6
TH, Galerkin	16	3.69e-2 (2.00)	6.13 (0.90)	9.55e-3 (2.33)	3.8
TH, Galerkin	32	8.77e-3 (2.07)	3.06 (1.00)	2.07e-3 (2.20)	1.9
TH, Galerkin	64	1.91e-3 (2.20)	1.40 (1.13)	4.89e-4 (2.08)	0.92

(E1) Spatial rates. Table 2 lists, for SV with CIP, the errors at T, the stabilization term $(\Delta t \sum_n S^{n+1}_{\mathrm{IMEX}}(e_h^{n+1}, e_h^{n+1}))^{1/2}$ with $e_h = u_h - s_h$, and the discrete $\ell^2(L^2)$ pressure error. Theorems 2 and 3 predict $h^{k+1/2}$ for the velocity and at least h^{k-1} for the pressure. For $k = 2$ the stabilization term converges with orders 2.63, 2.53, 2.49 and 2.50, the predicted $h^{k+1/2}$, and the L^2 velocity error with orders 2.50, 2.69, 2.84 and 2.91, at or above it. The pressure converges with orders between 3.14 and 2.53, declining towards the $h^{k+1/2}$ of the velocity and well above the proved h^{k-1} (Remark 5). The pressure sum contains the IMEX–Euler starting step, whose error is of order $\Delta t^{3/2}$; the last column omits it. The divergence $\|\nabla \cdot u_h\|_0$ of the SV velocity is below 10^{-13} in all runs. [Authors: the $k = 3$ runs are outstanding; the corresponding rows of Table 2 are left open.]

Table 2: Example 1, (E1): SV with CIP, $\nu = 10^{-6}$. Numbers in brackets: orders between consecutive meshes. "Stab." is $(\Delta t \sum_n S^{n+1}_{\mathrm{IMEX}}(e_h^{n+1}, e_h^{n+1}))^{1/2}$ with $e_h = u_h - s_h$; the last column is the pressure error summed from $n = 2$.

k	n	Δt	$\|u - u_h\|_0$	$\|u - u_h\|_1$	Stab.	$\|p - p_h\|_{\ell^2(L^2)}$	same, $n \geq 2$
2	4	5.00e-3	8.14e-2	2.11	0.514	2.23e-1	2.22e-1
2	8	5.00e-3	1.44e-2 (2.50)	0.730 (1.53)	8.32e-2 (2.63)	2.52e-2 (3.14)	2.51e-2 (3.14)
2	16	5.00e-3	2.22e-3 (2.69)	0.164 (2.15)	1.44e-2 (2.53)	3.47e-3 (2.86)	3.45e-3 (2.86)
2	32	5.00e-3	3.11e-4 (2.84)	3.54e-2 (2.21)	2.58e-3 (2.49)	5.27e-4 (2.72)	5.24e-4 (2.72)
2	64	5.00e-3	4.14e-5 (2.91)	8.45e-3 (2.07)	4.57e-4 (2.50)	9.10e-5 (2.53)	9.06e-5 (2.53)
3	4	–	–	–	–	–	–
3	8	–	–	–	–	–	–
3	16	–	–	–	–	–	–
3	32	–	–	–	–	–	–
3	64	–	–	–	–	–	–

Comparison of the discretizations. Table 3 and Figure 1 compare SV and TH, each with Galerkin and CIP, and TH also with grad-div, alone and together with CIP, on $n = 8, \ldots, 64$. SV

with CIP is the most accurate discretization on every mesh: at $n = 32$ its velocity error lies a factor 2.1 below TH with CIP and grad-div, 4.6 below TH with grad-div, and 4.9 below SV Galerkin, with the gap widening under refinement: at $n = 64$ the factors are 2.8, 9.1 and 8.3. CIP acts differently on the two pairs: for SV it reduces the H^1 error at $n = 32$ by a factor 22 (from 0.789 to $3.54 \cdot 10^{-2}$), while for TH alone the divergence error dominates ($\|\nabla \cdot u_h\|_0 \approx 1.9$ for Galerkin and 0.52 with CIP, against 10^{-2} with grad-div and 10^{-13} for SV), so TH needs grad-div before CIP pays. Figure 1 shows the full mesh sequences behind Table 3: the SV with CIP curves correspond to the runs of Table 2 (the figure plots the pressure error at T, the table its $\ell^2(L^2)$ sum), the unstabilized runs to those of Table 1.

Table 3: Example 1, $k = 2$, $\nu = 10^{-6}$: errors at T on $n = 32$ (numbers in brackets: orders from $n = 16$), velocity unknowns N_{dof} and wall-clock time per time step.

Method	N_{dof}	$\|u - u_h\|_0$	$\|u - u_h\|_1$	$\|p - p_h\|_0$	$\|\nabla \cdot u_h\|_0$	s/step
SV, Galerkin	49 280	1.53e-3 (1.99)	0.789 (1.14)	9.45e-4 (2.06)	7e-14	3.38
SV, CIP	49 280	3.11e-4 (2.84)	3.54e-2 (2.21)	7.45e-4 (2.72)	7e-14	8.09
TH, Galerkin	16 512	8.77e-3 (2.07)	3.06 (1.00)	2.07e-3 (2.20)	1.9	0.549
TH, CIP	16 512	1.99e-3 (2.14)	0.536 (1.09)	1.61e-3 (2.10)	0.52	2.00
TH, grad-div	16 512	1.42e-3 (1.78)	0.247 (0.94)	1.13e-3 (2.03)	1.0e-2	0.409
TH, CIP + grad-div	16 512	6.57e-4 (2.45)	5.55e-2 (1.73)	1.13e-3 (2.05)	8.2e-3	2.14

Convergence in time. At $\nu = 10^{-6}$ the vortex is quasi-steady and the temporal error is negligible; the time study therefore uses $\nu = 0.05$, for which $e^{-5\pi^2\nu t}$ decays by a factor of about 3.4 over $[0, T]$. On $n = 16$ we use $N = 10, 20, 40, 80$ and a reference run with $N = 320$ on the same mesh (Table 4, Figure 2). Both SV variants are second order, with velocity orders 2.10, 2.05 and 2.09 and pressure orders 2.15, 2.09 and 2.11 against the reference run; the stabilization leaves the temporal accuracy unchanged, the Galerkin and the CIP values agreeing to three digits.

Table 4: Example 1, $k = 2$, $n = 16$, $\nu = 0.05$: temporal error against the reference run with $N = 320$ on the same mesh, at the largest N, and the observed orders in N between consecutive runs.

Method	$\|u_h^N - u_h^{\mathrm{ref}}\|_0$, $N = 80$	orders	$\|p_h^N - p_h^{\mathrm{ref}}\|_0$, $N = 80$	orders
SV, Galerkin	2.48e-5	2.10, 2.05, 2.09	4.58e-6	2.15, 2.09, 2.11
SV, CIP	2.48e-5	2.10, 2.05, 2.09	4.58e-6	2.15, 2.09, 2.11

(E2) ν-sweep. Figure 3 shows $\|u(T) - u_h^N\|_0$ for $\nu = 10^{-1}, \ldots, 10^{-9}$, $k = 2$, $n = 16$, 32 and 64. For SV with CIP the error is bounded uniformly in ν and saturates at $2.2 \cdot 10^{-3}$, $3.1 \cdot 10^{-4}$ and $4.3 \cdot 10^{-5}$ for $\nu \lesssim 10^{-6}$, that is, with orders 2.83 and 2.87. SV Galerkin saturates at errors 2.8, 5.2 and 9.8 times larger, and its H^1 error (1.80, 0.89 and 0.45) is 11 to 53 times that of SV with CIP (0.164, $3.5 \cdot 10^{-2}$ and $8.5 \cdot 10^{-3}$). The SV saturation also lies below that of TH with grad-div ($4.90 \cdot 10^{-3}$, $1.45 \cdot 10^{-3}$ and $4.04 \cdot 10^{-4}$) by factors 2.2, 4.6 and 9.4, growing under refinement, while for $\nu \geq 10^{-2}$ the discretizations are comparable; the sharper separation appears in the pressure-robustness test (E3).

(E3) Pressure robustness. On $n = 32$ with $\nu = 10^{-6}$ we add the gradient force $R\nabla\phi$ with $R = 10^2, 10^4, 10^6$ and compare with the run for $R = 0$ (Table 5). The SV velocity error is the same

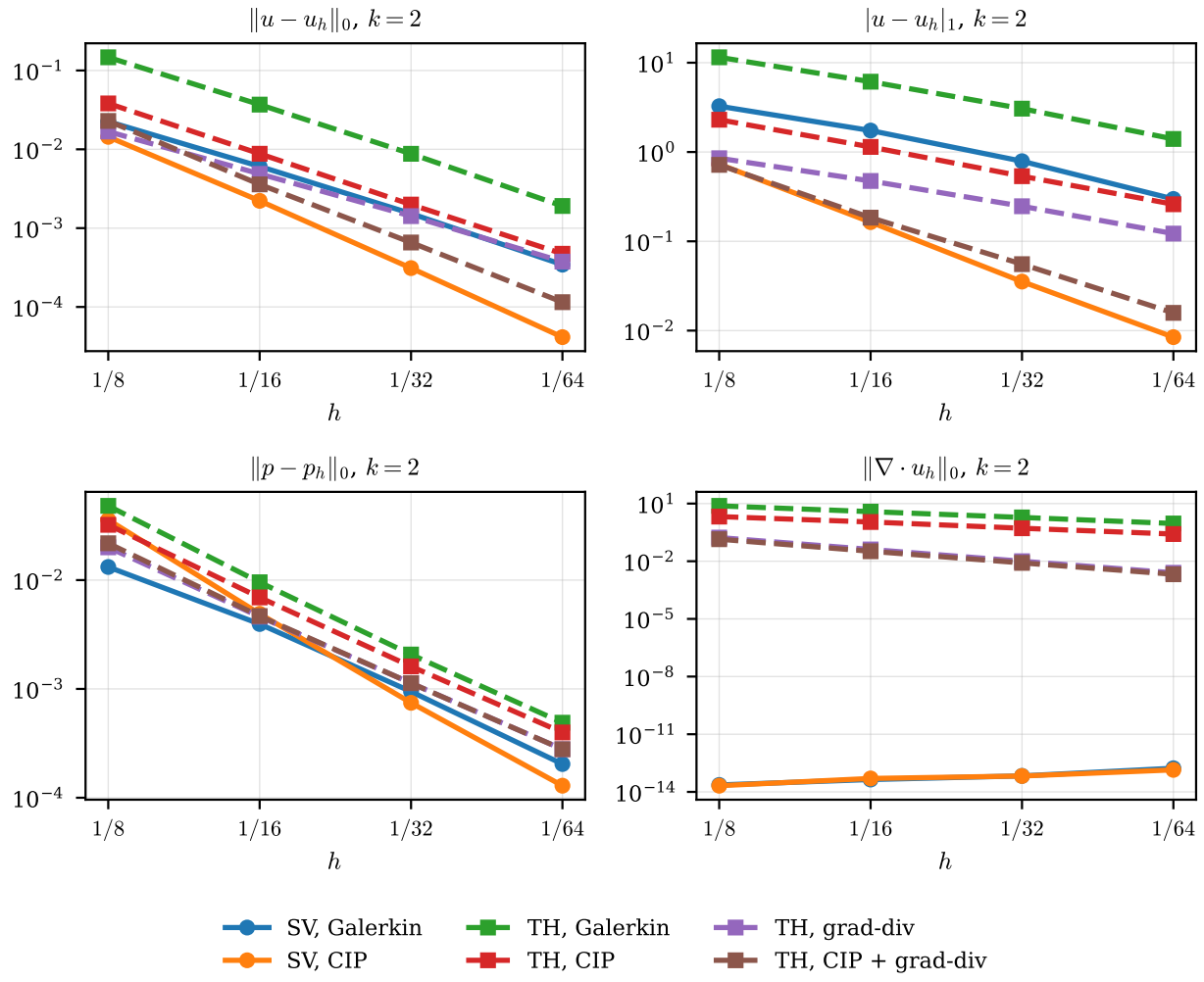


Figure 1: Example 1, $k = 2$, $\nu = 10^{-6}$: velocity and pressure errors and divergence norm at T against h (solid: SV; dashed: TH).

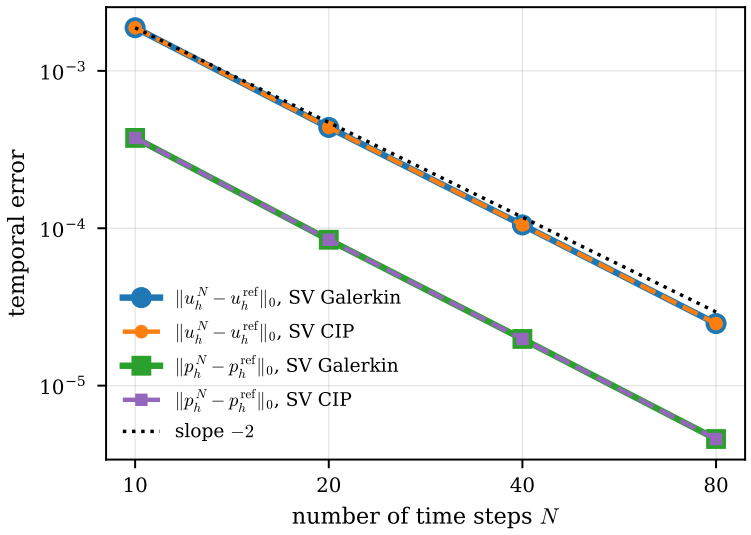


Figure 2: Example 1, $k = 2$, $n = 16$, $\nu = 0.05$: temporal error of velocity and pressure against the number of steps N. The dotted line has slope -2.

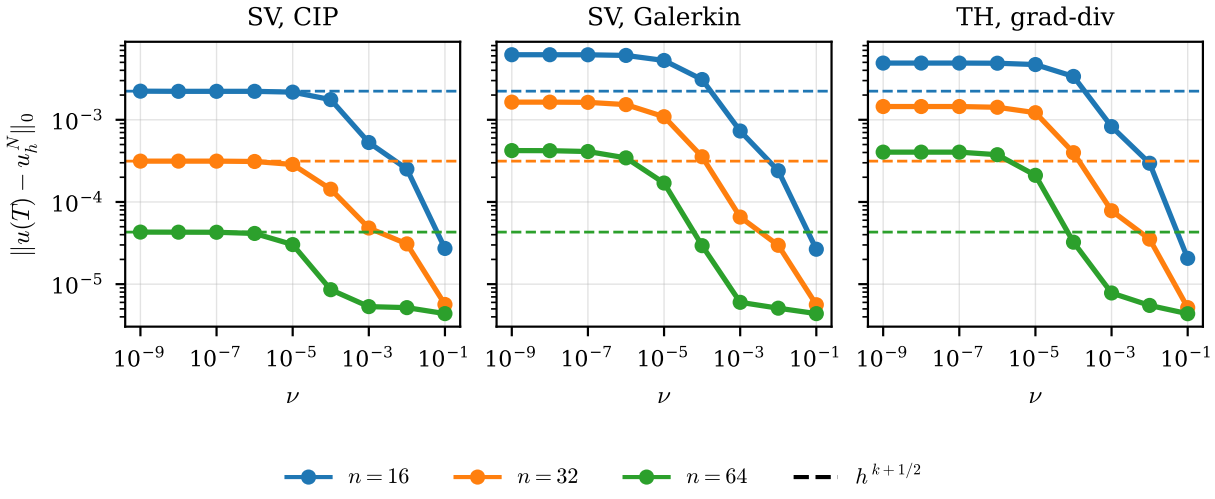


Figure 3: Example 1, (E2), $k = 2$, $T = 0.5$: $\|u(T) - u_h^N\|_0$ against ν on $n = 16$, 32 and 64, for SV with CIP, SV Galerkin and TH with grad-div. Dashed: $Ch^{k+1/2}$, fitted to SV with CIP on the same mesh.

to all digits shown for every R, and the remaining drift is of order $10^{-11}R$, the solver tolerance applied to a right-hand side of size R; the SV pressure error grows in proportion to $\|p + R\phi\|_0$, as it must, while the velocity does not see the pressure. Every TH variant changes in proportion to R or diverges: Galerkin diverges already at $R = 10^2$, the stabilized variants carry drifts of order one at $R = 10^2$ and of order ten at $R = 10^4$, and only CIP with grad-div completes the run at $R = 10^6$, with a drift of $2.6 \cdot 10^2$.

Table 5: Example 1, (E3), $n = 32$, $\nu = 10^{-6}$, $k = 2$: drift $\max |u_h^R - u_h^0|$ at T and the velocity error. A dash means that the run diverged.

Method	$R = 10^2$	$R = 10^4$	$R = 10^6$	$\|u - u_h\|_0$, $R = 0 \to 10^6$
SV, Galerkin	6.06e-9	2.30e-7	1.16e-5	1.53e-3 for all R
SV, CIP	6.87e-9	1.08e-6	5.31e-6	3.11e-4 for all R
TH, Galerkin	–	–	–	8.77e-3 at $R = 0$
TH, CIP	1.84	13.0	–	1.99e-3 to 4.03
TH, grad-div	0.547	7.11	–	1.42e-3 to 1.87
TH, CIP + grad-div	0.144	2.40	2.56e2	6.57e-4 to 105

(E4) Role of the three CIP terms. The analysis needs S_2 and S_3 only for the recovery step. To see whether they matter in practice, and whether the rate of Theorem 2 survives without them, we run SV with S_1 only, $S_1 + S_2$ and $S_1 + S_2 + S_3$, each with $(\delta_1, \delta_2, \delta_3) = s\,(10^{-1}, 10^{-2}, 10^{-3})$, $s \in \{0.1, 1, 10\}$, on $n = 8, \dots, 64$ (Table 6). With these weights S_3 is much weaker than the other terms, so we add $S_1 + S_2 + S_3$ with $\delta_3 = \delta_1$, which raises the third term to the weight of the first. The table also lists the TV ratio $\mathrm{TV}(\omega_h)/\mathrm{TV}(\omega)$ of the vorticity on a 128^2 grid, the assembly time per step, and the number of nonzeros of the system matrix.

A geometric scaling $\delta_{j+1}/\delta_j = \beta$ is meant to give the three terms similar weight: each further derivative of a discrete function costs a factor h^{-1}, which the extra h_F^2 in S_{j+1} cancels, and the factor β compensates the larger constants of the higher derivatives; this example uses $\beta = 10^{-1}$, the shear-layer runs of Table 12 use $\beta = 10^{-2}$. Table 7 checks the choice with the computed values of the terms with unit weights at T: a base β is balanced when $S_{j+1}/S_j \approx \beta^{-1}$. All three terms couple only the degrees of freedom of the two elements sharing a face, so the stencil of the system matrix stays local; in the shear-layer runs of Table 12, activating S_2 and S_3 fills in about 9% more nonzeros within these blocks than S_1 alone (340 968 against 313 320 at $n = 16$), and S_3 adds none beyond S_2. In the double shear layer (Section 5.3, $n = 16$, Table 12), S_1 alone gives TV ratios of 2.00, 1.11 and 0.73 for $s = 0.1, 1, 10$, against 1.87, 0.96 and 0.60 with all three terms, and $S_1 + S_2$ and $S_1 + S_2 + S_3$ agree to 0.01 in every solution quantity; at $s = 1$ and $s = 10$, S_1 removes the over- and undershoots on its own, while at $s = 0.1$ a vorticity undershoot of about 0.1 remains for every combination of the terms.

On Example 1 (Table 6), S_1 alone is at least as accurate as the full form and converges with the same or higher orders: at $s = 1$ and $n = 64$, $2.48 \cdot 10^{-5}$ (3.14) against $4.14 \cdot 10^{-5}$ (2.91), with the stabilization term of both at the predicted order 2.5. The reweighted third term ($\delta_3 = \delta_1$, a hundredfold increase) stays within a factor of about two of the full form in either direction, so S_3 remains numerically inactive at every tested weight; its role is the recovery step in the proof. The measured ratios in Table 7 explain this: for the discrete error e_h, S_2/S_1 reaches 172–188 at $s \geq 1$, well above the balanced value 10 of the base used here, so the second term carries 17–19 times the weighted budget of the first, while S_3/S_2 stays between 0.03 and 0.24, so with $\delta_3 = 10^{-3}$ the third term contributes below one percent of the stabilization budget. The number of nonzeros grows by

5% from S_1 to $S_1 + S_2$ and not at all from S_2 to S_3, as in the shear-layer runs (Table 12), and assembling the full form costs about 4.2 times S_1 alone (18 s against 4.3 s per step at $n = 64$).

Table 6: Example 1, (E4), SV, $k = 2$, $\nu = 10^{-6}$, $n = 64$ (numbers in brackets: orders from $n = 32$): CIP terms and strength s, $(\delta_1, \delta_2, \delta_3) = s\,(10^{-1}, 10^{-2}, 10^{-3})$ for the active terms. "Stab." as in Table 2; TV ratio of the vorticity on a 128^2 grid; assembly time per step of the convection and CIP matrices.

Terms	s	$\|u - u_h\|_0$	$\|u - u_h\|_1$	Stab.	TV ratio	ms/step	nonzeros
Galerkin	–	3.44e-4 (2.16)	0.298 (1.41)	–	2.02	1 565	5 670 644
S_1	0.1	1.65e-4 (2.21)	3.65e-2 (1.40)	1.05e-4 (2.22)	1.02	4 563	10 327 412
$S_1 + S_2$	0.1	1.39e-4 (2.36)	3.06e-2 (1.63)	1.72e-4 (2.43)	1.01	7 784	10 864 628
$S_1 + S_2 + S_3$	0.1	1.40e-4 (2.37)	3.07e-2 (1.60)	1.72e-4 (2.43)	1.01	18 194	10 864 628
$S_1 + S_2 + S_3$, $\delta_3 = \delta_1$	0.1	1.50e-4 (2.45)	3.34e-2 (1.59)	1.91e-4 (2.70)	1.01	18 062	10 864 628
S_1	1	2.48e-5 (3.14)	8.00e-3 (1.98)	1.03e-4 (2.48)	1.00	4 309	10 327 412
$S_1 + S_2$	1	4.20e-5 (2.91)	8.59e-3 (2.18)	4.57e-4 (2.49)	1.00	7 535	10 864 628
$S_1 + S_2 + S_3$	1	4.14e-5 (2.91)	8.45e-3 (2.07)	4.57e-4 (2.50)	1.00	18 070	10 864 628
$S_1 + S_2 + S_3$, $\delta_3 = \delta_1$	1	3.39e-5 (3.22)	7.54e-3 (2.35)	5.09e-4 (2.82)	1.00	18 276	10 864 628
S_1	10	5.02e-5 (2.70)	1.09e-2 (1.96)	2.78e-4 (2.55)	1.00	4 559	10 327 412
$S_1 + S_2$	10	1.42e-4 (2.86)	2.05e-2 (2.15)	1.44e-3 (2.50)	1.00	7 744	10 864 628
$S_1 + S_2 + S_3$	10	1.33e-4 (2.86)	1.89e-2 (2.07)	1.44e-3 (2.50)	1.00	17 442	10 864 628
$S_1 + S_2 + S_3$, $\delta_3 = \delta_1$	10	2.51e-4 (3.43)	7.76e-2 (1.44)	2.43e-3 (2.94)	1.03	18 311	10 864 628

Table 7: Example 1, (E4), $n = 64$: ratios of the CIP terms $S_j(w, w)$ with unit weights at T, for $w = u_h$ and $w = e_h = u_h - s_h$, in the run with all three terms.

s	S_2/S_1 $(w = u_h)$	S_3/S_2 $(w = u_h)$	S_2/S_1 $(w = e_h)$	S_3/S_2 $(w = e_h)$
0.1	5.3	0.24	24.7	0.08
1	36.2	0.03	172	0.03
10	46.0	0.03	188	0.03

(E5) Time-step restriction. With SV and CIP on $n = 32$ we vary $\Delta t/h \in \{1/8, 1/4, 1/2, 1, 2, 4\}$ and compare three normalizations of the stabilization: the original $\|\tilde{u}_h^{n+1}\|_{0,\infty}^{-1}$ of (8), the one-sided $\max(10^{-2}, \|\tilde{u}_h^{n+1}\|_{0,\infty})^{-1}$ used elsewhere in this section, and the clamped N_{n+1}^{-1} of Remark 4 with $u_{\min}$ half of $\min_t \|u(t)\|_{0,\infty}$ and $U_{\max} = 1.25 \max_t \|u(t)\|_{0,\infty}$ (Table 8). A run counts as unbounded when $\|u_h^n\|_0$ exceeds $10\|u(t_n)\|_0$. The last column shows the range of $\|\tilde{u}_h^{n+1}\|_{0,\infty}$; the clamp acts only where this range leaves $[u_{\min}, U_{\max}]$. No restriction is visible: every run is stable up to $\Delta t = 4h$, and the three normalizations give identical errors to all digits shown. The error stays at the spatial level $3.11 \cdot 10^{-4}$ of Table 2 for $\Delta t \le h$ and even drops slightly for the largest steps ($2.20 \cdot 10^{-4}$ at $\Delta t = 4h$), a cancellation within the spatial error level: the solution is quasi-steady at this ν, so the $O(\Delta t^2)$ part is negligible even for the largest step. The range of $\|\tilde{u}_h^{n+1}\|_{0,\infty}$ is $[2.00, 2.00]$ throughout — the exact value $\|u(t)\|_{0,\infty} = 2$ reproduced to three digits — so it never leaves $[u_{\min}, U_{\max}]$ and the clamps are never active.

Table 8: Example 1, (E5), SV with CIP, $k = 2$, $n = 32$, $\nu = 10^{-6}$: $\|u(T) - u_h^N\|_0$ for the three normalizations of the stabilization.

$\Delta t/h$	original	one-sided	clamped	range of $\|\tilde{u}_h^{n+1}\|_{0,\infty}$
1/8	3.11e-4	3.11e-4	3.11e-4	[2.00, 2.00]
1/4	3.11e-4	3.11e-4	3.11e-4	[2.00, 2.00]
1/2	3.11e-4	3.11e-4	3.11e-4	[2.00, 2.00]
1	3.08e-4	3.08e-4	3.08e-4	[2.00, 2.00]
2	2.85e-4	2.85e-4	2.85e-4	[2.00, 2.00]
4	2.20e-4	2.20e-4	2.20e-4	[2.00, 2.00]

5.2 Three-dimensional travelling vortex (a variant of the analysed scheme)

The vortex benchmark starts from a radially symmetric swirling flow, which is a steady solution of the incompressible Euler equations [24]. By Galilean invariance, the vortex can be carried along by a uniform background velocity U_0, as in Gresho's vortex [25]. To obtain a Navier–Stokes solution, we move the viscous term into the forcing, $f = -\nu\Delta u$; the velocity–pressure pair is then an exact solution for every ν.

In all cases the vortex centre starts at $c_0 = \frac{1}{2}\mathbf{1}$, the centre of the periodic domain, and moves with constant velocity U_0. Relative to the moving centre we use the coordinates

$$\xi_1 = x - \tfrac{1}{2} - U_{0,1}\,t, \qquad \xi_2 = y - \tfrac{1}{2} - U_{0,2}\,t, \qquad r^2 = \xi_1^2 + \xi_2^2, \tag{44}$$

where each ξ_i is taken as its nearest periodic image in $[-\frac{1}{2}, \frac{1}{2})$. The vortex axis is parallel to the z-axis, and the axial coordinate enters only through the transport. For a radial profile g, the swirl $A\,g(r^2)(-\xi_2, \xi_1)$ is balanced by the pressure gradient $\partial_r p = A^2 r\,g(r^2)^2$, which gives an Euler solution; adding the viscous forcing above turns it into a Navier–Stokes solution. The amplitude A is chosen so that the peak swirl speed is one. We take $\nu = 10^{-7}$ and integrate up to $T = 0.5$. The vortex is transported obliquely across the mesh, convection is of order one, and the pressure is not polynomial, so the test is demanding for all discretizations. The domain is the periodic unit cube, so the no-slip condition of the analysis does not hold; the results illustrate the behaviour of the method rather than test the theorems.

We stress that the three-dimensional computations in this subsection *do not use the scheme analysed in this paper*: convection is treated fully explicitly, and the stabilization is a single gradient-jump CIP term rather than the three-term form S (see below). Theorems 1–3 therefore do not apply. We include the results because they show pressure robustness and temporal accuracy of SV versus TH in three dimensions; we draw no conclusions about spatial convergence rates from the two SV meshes.

The three-dimensional test uses a columnar vortex with an axial jet of Batchelor type. Its profile is $q = \max\{0, 1 - r^2/R^2\}$ with $R = 0.45$, $m = 5$ and $J = \frac{1}{2}$, the transport velocity is $U_0 = (1, \frac{1}{2}, \frac{1}{4})^\top$, and the amplitude is $A = \big(R\,\sigma^{1/2}(1-\sigma)^m\big)^{-1} \approx 11.87$ with $\sigma = 1/(2m+1)$:

$$\begin{aligned} u(x,y,z,t) &= \begin{pmatrix} 1 - A\,\xi_2\,q^5 \\ \frac{1}{2} + A\,\xi_1\,q^5 \\ \frac{1}{4} + J\,q^5 \end{pmatrix}, \qquad p(x,y,z,t) = -\frac{A^2R^2}{22}\,q^{11}, \\ f(x,y,z,t) &= -\frac{20\,\nu\,q^3}{R^2} \begin{pmatrix} -2A\,(2-3q)\,\xi_2 \\ 2A\,(2-3q)\,\xi_1 \\ J\,(4-5q) \end{pmatrix}. \end{aligned} \tag{45}$$

Table 9: Travelling vortex, $d = 3$: errors at time T (numbers in brackets: convergence order with respect to the previous mesh). A dash (–) indicates that the computation diverged.

Method	n	dofs	$\|u - u_h\|_0$	$\|u - u_h\|_1$	$\|p - p_h\|_0$
SV P_3, Galerkin	3	9 477	–	–	–
	4	22 464	2.0e-1	26.7	2.1e-1
SV P_3, CIP 10^{-3}	3	9 477	1.1e-1 (1.65)	2.13 (1.07)	1.0e-1 (0.99)
	4	22 464	4.9e-2 (2.73)	1.35 (1.59)	4.5e-2 (2.94)
TH P_2, Galerkin	8	12 288	2.1e-1 (1.73)	9.80 (0.19)	1.2e-1 (1.35)
TH P_2, grad-div	8	12 288	4.2e-2 (2.47)	1.31 (1.20)	4.5e-2 (2.28)

Since the profile vanishes for $r \geq R$, the field is exactly periodic. It belongs to $H^{m+1/2-\varepsilon} = H^{5.5-\varepsilon}$, and all three velocity components are nonzero.

For SV we use P_3/P_2^{disc}, the lowest admissible degree in three dimensions ($k = d = 3$), on Alfeld splits of Kuhn triangulations with $n = 2, 3, 4$. TH uses P_2/P_1 on Kuhn meshes with $n = 2, 4, 6, 8$.

The three-dimensional implementation differs from the analysed scheme (8) in two ways:

- Convection is treated *fully explicitly*: $((\tilde{u}_h^{n+1} \cdot \nabla)\tilde{u}_h^{n+1}, v_h)$, so the system matrix remains constant in time and no nonlinear solve is needed per step.
- The CIP stabilization uses *gradient jumps* across interior facets: $\delta \sum_F h_F^2 \int_F [\![\nabla u_h]\!] : [\![\nabla v_h]\!] \, \mathrm{d}s$, instead of the three-term form S.

With explicit convection, Theorem 1 no longer applies, and the time step is subject to a CFL condition.

Table 9 and Figures 4–6 summarize the results.

Unstabilized SV diverges on the mesh $n = 3$, and on $n = 4$ its H^1 error is 26.7; with explicit convection, Galerkin SV is simply not usable at this viscosity. SV with CIP ($\delta = 10^{-3}$), on the other hand, stays stable on all meshes we tried. From $n = 3$ to $n = 4$ its observed orders are 2.73 in L^2, 1.59 in H^1 and 2.94 for the pressure, but two coarse meshes are not enough to say anything about asymptotic rates, and Theorem 2 does not cover this variant anyway. Unstabilized TH also does poorly: $|u - u_h|_1 \approx 10$ and $\|\nabla \cdot u_h\|_0 \approx 6$ stay large. The best TH results come from grad-div, which reaches an accuracy per degree of freedom comparable to SV with CIP at this moderate pressure.

For the temporal tests we use SV on $n = 3$ and TH on $n = 6$, with $N = 40, 80, 160, 320$ and a reference run with $N = 1280$. All stabilized methods reach observed orders between 2.0 and 2.15 for velocity and pressure, so BDF2 keeps its accuracy in three dimensions as well. The pressure-robustness test (SV on $n = 3$, TH on $n = 6$) again separates the pairs. The SV velocity changes by at most 2×10^{-10}, even for $R = 10^6$, where $\|p\|_0 \approx 5.6 \times 10^4$. For TH the change grows with R: with CIP and grad-div, $\max |u_h^R - u_h^0|$ increases from 6×10^{-3} at $R = 1$ to 0.65 at $R = 10^2$, and every TH run diverges for $R \geq 10^4$.

Overall, the three-dimensional results agree with those of Example 1 (Section 5.1). They do not, however, show the rate $h^{k+1/2}$: the meshes needed to reach the asymptotic regime are beyond what our direct solvers can handle. Nor do they test the recovery property (11), on which the three-dimensional velocity estimate depends (Remark 3).

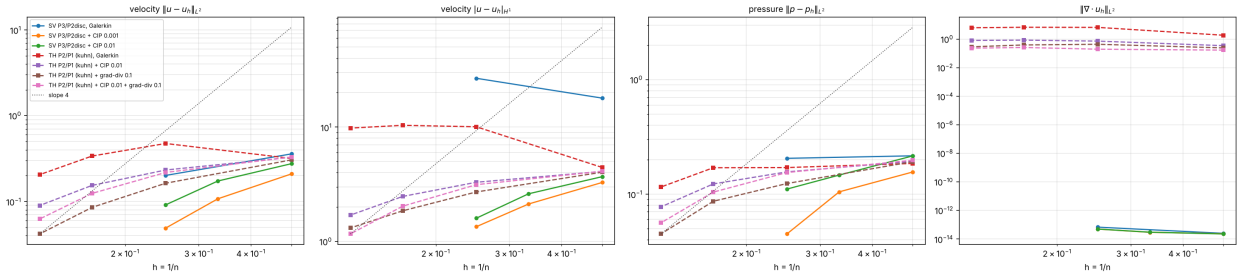


Figure 4: Travelling vortex, $d = 3$: errors at time T against h (solid lines: SV P_3; dashed: TH P_2). The SV Galerkin run on $n = 3$ is absent because it diverged.

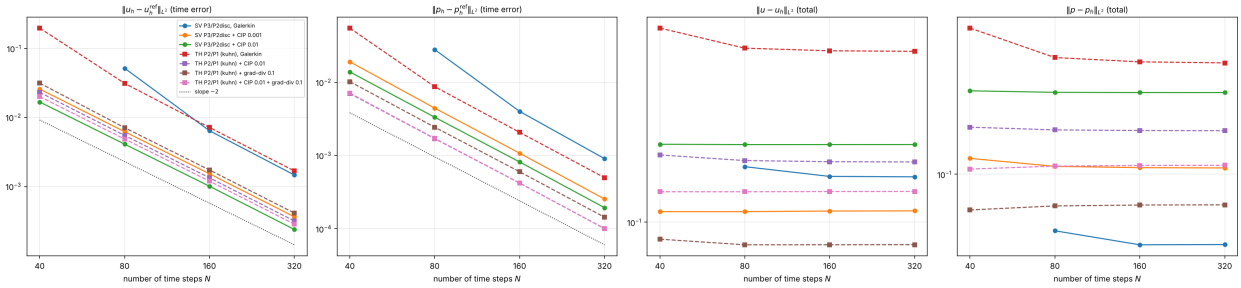


Figure 5: Travelling vortex, $d = 3$: time-discretization convergence against the number of steps N (SV at $n = 3$, TH at $n = 6$; temporal error measured as the difference from the $N = 1280$ reference run). The dotted line shows slope -2.

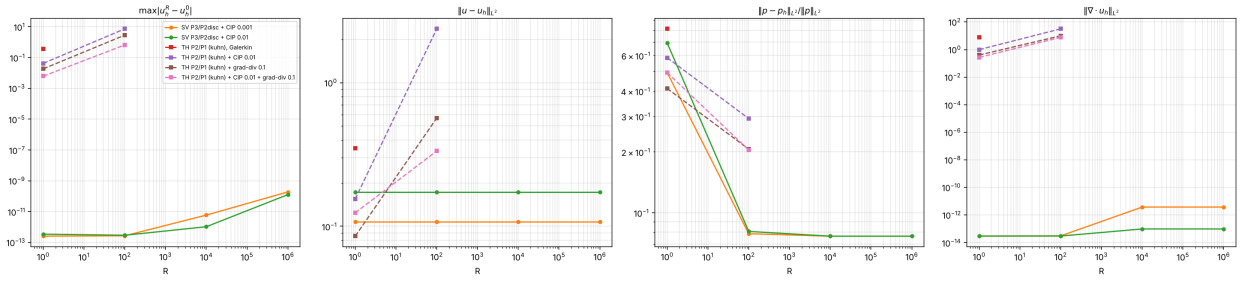


Figure 6: Travelling vortex, $d = 3$: pressure robustness (SV at $n = 3$, TH at $n = 6$). Missing points indicate divergence.

5.3 Spurious oscillations in an under-resolved double shear layer

Our last test is the doubly periodic shear layer of Bell, Colella and Glaz [26], a standard benchmark for spotting spurious vortical structures in under-resolved simulations [27, 28]. On $(0,1)^2$ the initial velocity consists of two thin shear layers of opposite direction, perturbed by a small sinusoidal mode:

$$u_0(x) = \begin{pmatrix} \tanh\big(\rho\,(\tfrac{1}{4} - |x_2 - \tfrac{1}{2}|)\big) \\ \delta \sin\big(2\pi(x_1 + \tfrac{1}{4})\big) \end{pmatrix}. \tag{46}$$

We take $\rho = 80$, $\delta = 0.05$ and $\nu = 10^{-4}$. The layers roll up into two large vortices connected by thin braids. On coarse meshes, unstabilized methods tend to produce spurious secondary vortices along the braids, a well-known artefact that makes this problem a good diagnostic [27].

The reference solution is a fully dealiased pseudo-spectral computation with 512^2 Fourier modes. We compare SV and TH with $k = 2$ on the coarse mesh $n = 16$ at $t = 1$, and repeat the comparison on $n = 32$. Unlike the other tests, the time step here follows a CFL condition, $\Delta t \le 0.2/(n \max |u_h^0|)$, rounded so that the output times are multiples of Δt: $\Delta t = 10^{-2}$ on $n = 16$ and $\Delta t = 1/170 \approx 5.9 \cdot 10^{-3}$ on $n = 32$. Oscillations are quantified in two ways: by the overshoots and undershoots of the discrete vorticity ω_h beyond the reference range, normalized by $\max |\omega_{\mathrm{ref}}|$, and by the total variation ratio $\mathrm{TV}(\omega_h)/\mathrm{TV}(\omega_{\mathrm{ref}})$. A ratio above one means excess variation, a ratio below one means that the solution is smoother than the reference.

Unstabilized SV shows strong element-scale oscillations, with a total variation ratio of 4.3 and a vorticity undershoot of 45%. Unstabilized TH is worse still: its total variation ratio is 6.3, its divergence norm is $\|\nabla \cdot u_h\|_0 = 41$, and the velocity overshoots by 0.7. For SV, CIP with unit strength ($s = 1$) brings the total variation ratio down to 0.96 and removes the over- and undershoots. At a tenth of this strength ($s = 0.1$) oscillations are still visible, and much stronger CIP smears the vortex cores. For TH, grad-div matters as well: with CIP alone, velocity overshoots of 0.12 remain, whereas grad-div alone removes the over- and undershoots and gives a ratio of 1.10. Using both lowers the ratio to 0.74.

Even with the oscillations gone, the relative vorticity error stays between 0.74 and 0.95 for all stabilized runs. This error is due to the poor resolution of the thin braids on the mesh $n = 16$, not to oscillations; no stabilization can recover scales that the mesh cannot represent. On $n = 32$ (Table 11) the ranking is the same and the stabilized vorticity errors drop only to 0.66–0.74, so the braids are still under-resolved. Two details differ: CIP with $s = 0.1$ is now the most accurate SV run (0.66 against 0.69 for $s = 1$), and grad-div alone leaves a TV ratio of 1.47 for TH. We note that the CIP strength $s = 1$ in Figure 7 was chosen afterwards, by comparing vorticity errors with the reference; it is a best case, not a recommended parameter value.

Table 12 isolates the contribution of the three CIP terms on the mesh $n = 16$, for the strengths of experiment (E4) in Section 5.1. At $s = 1$ and $s = 10$, S_1 alone removes the over- and undershoots, and at unit strength it brings the TV ratio from 4.30 down to 1.11; adding S_2 lowers it to 0.97, and S_3 changes no solution quantity by more than 0.01. The price of the higher-order terms is assembly time: with the transport field, and hence all matrices, changing in every step, S_1 takes 52 ms per step to assemble, $S_1 + S_2$ takes 92 ms and the full form 221 ms, against 17 ms for the Galerkin method. The stabilization is also dissipative: over $[0, 1.5]$ the CIP terms account for about 21% of the kinetic-energy loss at $s = 0.1$, 35% at $s = 1$ and 57% at $s = 10$, with the viscous term contributing most of the rest. This over-dissipation at large s is what smears the vortex cores, and it shows in the relative L^2 velocity error, which grows from 0.144 (Galerkin) to 0.258 ($s = 1$) and 0.435 ($s = 10$) while the vorticity error improves.

Table 10: Double shear layer, $k = 2$, $n = 16$, $t = 1$. CIP strength s means $(\delta_1, \delta_2, \delta_3) = s\,(10^{-2}, 10^{-4}, 10^{-6})$. The column labelled "error" is the relative L^2 error of ω_h against the spectral reference.

Method	TV ratio	ω over / under	u over / under	error	$\|\nabla \cdot u_h\|_0$
SV, Galerkin	4.30	0.18 / 0.45	0 / 0	1.05	2e-10
SV, CIP 0.1	1.87	0 / 0.10	0 / 0	0.76	2e-10
SV, CIP 1	0.96	0 / 0	0 / 0	0.74	2e-10
TH, Galerkin	6.31	1.10 / 1.29	0.70 / 0.70	1.88	41
TH, CIP 1	1.50	0 / 0.09	0.12 / 0.12	0.95	11
TH, grad-div	1.10	0 / 0	0 / 0	0.82	0.46
TH, CIP 1 + grad-div	0.74	0 / 0	0 / 0	0.79	0.46

Table 11: Double shear layer, $k = 2$, $n = 32$, $t = 1$; columns as in Table 10.

Method	TV ratio	ω over / under	u over / under	error	$\|\nabla \cdot u_h\|_0$
SV, Galerkin	2.49	0.20 / 0.38	0 / 0	0.74	9e-10
SV, CIP 0.1	1.52	0.17 / 0.25	0 / 0	0.66	9e-10
SV, CIP 1	1.02	0.01 / 0.07	0 / 0	0.69	9e-10
TH, Galerkin	3.01	0.83 / 0.33	0 / 0	0.88	7.2
TH, CIP 1	1.21	0.44 / 0.01	0 / 0	0.73	5.0
TH, grad-div	1.47	0 / 0	0 / 0	0.73	0.20
TH, CIP 1 + grad-div	1.01	0 / 0	0 / 0	0.74	0.16

Table 12: Double shear layer, SV, $k = 2$, $n = 16$, $t = 1$: role of the three CIP terms, experiment (E4) of Section 5.1. CIP strength s as in Table 10; "ω error" and "u error" are the relative L^2 errors against the spectral reference; "assembly" is the time per step for the convection and CIP matrices, "step" the total time per step, and "nnz" the number of nonzeros of the system matrix.

Terms	s	TV ratio	ω over / under	ω error	u error	assembly	step	nnz
Galerkin	–	4.30	0.18 / 0.45	1.05	0.144	17 ms	246 ms	178 152
S_1	0.1	2.00	0 / 0.11	0.77	0.174	52 ms	316 ms	313 320
$S_1 + S_2$	0.1	1.88	0 / 0.10	0.76	0.176	93 ms	334 ms	340 968
$S_1 + S_2 + S_3$	0.1	1.87	0 / 0.10	0.76	0.176	221 ms	456 ms	340 968
S_1	1	1.11	0 / 0	0.75	0.265	52 ms	380 ms	313 320
$S_1 + S_2$	1	0.97	0 / 0	0.74	0.259	92 ms	385 ms	340 968
$S_1 + S_2 + S_3$	1	0.96	0 / 0	0.74	0.258	221 ms	515 ms	340 968
S_1	10	0.73	0 / 0	0.81	0.383	51 ms	499 ms	313 320
$S_1 + S_2$	10	0.61	0 / 0	0.84	0.435	92 ms	481 ms	340 968
$S_1 + S_2 + S_3$	10	0.60	0 / 0	0.84	0.435	226 ms	608 ms	340 968

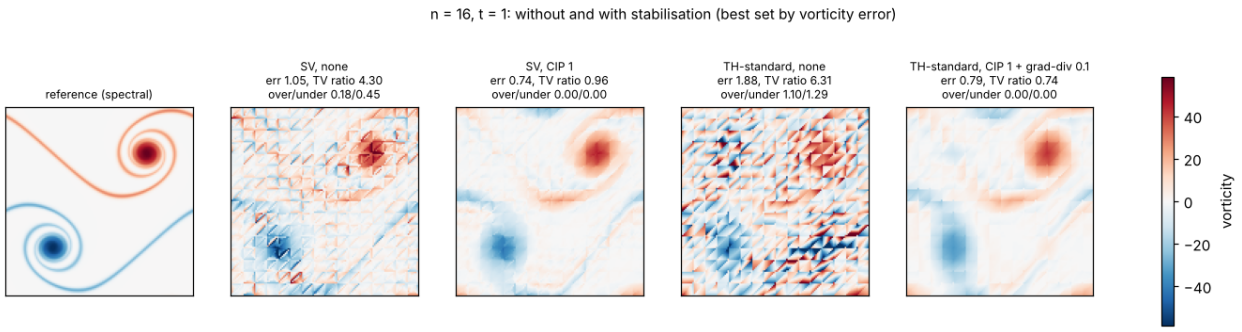


Figure 7: Double shear layer: vorticity fields at $t = 1$ on mesh $n = 16$, displayed on a common colour scale. From left to right: spectral reference, SV Galerkin, SV with the best CIP ($s = 1$), TH Galerkin, and TH with the best stabilization (CIP $s = 1$ and grad-div).

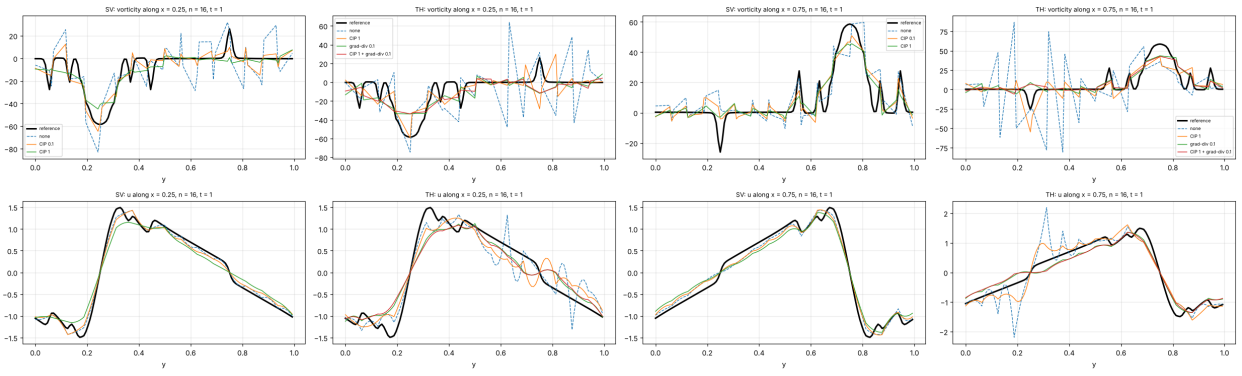


Figure 8: Double shear layer: one-dimensional profiles of ω and u_1 along the vertical lines $x_1 = 0.25$ and $x_1 = 0.75$ at $t = 1$, $n = 16$.

A Discrete velocity time-derivative estimate

This appendix explains the auxiliary estimate used in the pressure bound, namely

$$\Delta t \sum_{n=1}^{N-1} \|D_t^{\mathrm{BDF2}} e_u^{n+1}\|_{-1,h}^2 \le C(h^{2k} + \Delta t^4), \tag{47}$$

which, up to the projection residual ρ, is the bound discussed in Remark 5. We do not prove (47). Lemma 5 proves its divergence-free projected version, and Proposition 1 derives from it a bound with an additional factor h^{-2}, which gives Theorem 3(iii). Here $e_u^{n+1} = u^{n+1} - u_h^{n+1}$ denotes the total velocity error and $\|g\|_{-1,h} = \sup_{0\neq v_h\in V_h}(g, v_h)/\|\nabla v_h\|_0$. We also use the projected divergence-free dual norm $\|g\|_{-1,h,\mathrm{div}} = \sup_{0\neq v_h\in V_h^{\mathrm{div}}}(g, v_h)/\|\nabla v_h\|_0$.

Subtract the fully discrete momentum equation in (8) from the exact momentum equation in (2) at t_{n+1}. After adding and subtracting the BDF2 derivative of the exact solution, one obtains, for every $v_h \in V_h$,

$$(D_t^{\mathrm{BDF2}} e_u^{n+1}, v_h) = (\tau^{n+1}, v_h) - \nu(\nabla e_u^{n+1}, \nabla v_h) + (p^{n+1} - p_h^{n+1}, \nabla\cdot v_h) - C^{n+1}(v_h) + S_{\mathrm{IMEX}}^{n+1}(u_h^{n+1}, v_h), \tag{48}$$

where

$$C^{n+1}(v_h) = (u^{n+1}\cdot\nabla u^{n+1}, v_h) - ((\tilde{u}_h^{n+1}\cdot\nabla)u_h^{n+1}, v_h), \qquad \tau^{n+1} = D_t^{\mathrm{BDF2}} u^{n+1} - \partial_t u(t_{n+1}).$$

The pressure term is the obstruction to a completely pressure-robust estimate of the full norm $\|D_t^{\mathrm{BDF2}} e_u^{n+1}\|_{-1,h}$, because the supremum is taken over all of V_h, not only over the discretely divergence-free subspace. Indeed,

$$|(p^{n+1} - p_h^{n+1}, \nabla\cdot v_h)| \le \sqrt{d}\, \|p^{n+1} - p_h^{n+1}\|_0 \|\nabla v_h\|_0,$$

so the full dual norm necessarily contains a pressure-error contribution. However, this obstruction disappears after projection onto V_h^{div}.

The BDF2 consistency error has the Peano-kernel representation

$$\tau^{n+1} = \frac{1}{2\Delta t}\int_{t_{n-1}}^{t_{n+1}} K_n(s)\,\partial_{ttt}u(s)\,\mathrm{d}s, \qquad |K_n(s)| \le C\Delta t^2.$$

Thus, if $\partial_{ttt}u \in L^2(0,T;H^{-1})$,

$$\Delta t \sum_{n=1}^{N-1} \|\tau^{n+1}\|_{H^{-1}}^2 \le C\Delta t^4 \|\partial_{ttt}u\|_{L^2(0,T;H^{-1})}^2. \tag{49}$$

The viscous part is controlled by the velocity estimate together with the projection estimate for $\nabla\eta$:

$$\Delta t \sum_{n=1}^{N-1} \nu^2 \|\nabla e_u^{n+1}\|_0^2 \le C\Delta t \sum_{n=1}^{N-1} \left(\nu^2\|\nabla e_h^{n+1}\|_0^2 + \nu^2\|\nabla\eta^{n+1}\|_0^2\right) \le C(h^{2k} + \Delta t^4). \tag{50}$$

The convective and stabilization residuals satisfy, as in the proof of Theorem 3,

$$\Delta t \sum_{n=1}^{N-1} \|R_{\mathrm{conv}}^{n+1}\|_{-1,h}^2 + \Delta t \sum_{n=1}^{N-1} \|R_{\mathrm{stab}}^{n+1}\|_{-1,h}^2 \le C(h^{2k} + \Delta t^4). \tag{51}$$

Lemma 5 (Projected time-derivative estimate)**.** *The discrete time derivative of the velocity error satisfies*

$$\Delta t \sum_{n=1}^{N-1} \|D_t^{\mathrm{BDF2}} e_u^{n+1}\|_{-1,h,\mathrm{div}}^2 \le C(h^{2k} + \Delta t^4). \tag{52}$$

Proof. Restrict (48) to $v_h \in V_h^{\mathrm{div}}$. Then $(p^{n+1} - p_h^{n+1}, \nabla \cdot v_h) = 0$, so the pressure residual vanishes and

$$\frac{|(D_t^{\mathrm{BDF2}} e_u^{n+1}, v_h)|}{\|\nabla v_h\|_0} \le C\|\tau^{n+1}\|_{H^{-1}} + C\nu\|\nabla e_u^{n+1}\|_0 + C\|R_{\mathrm{conv}}^{n+1}\|_{-1,h} + C\|R_{\mathrm{stab}}^{n+1}\|_{-1,h}.$$

Taking the supremum over V_h^{div}, squaring, multiplying by Δt, summing over n, and applying (49), (50), and (51) proves (52). □

The preceding lemma is pressure-robust, but it controls only the divergence-free component of the dual norm.

Proposition 1. *Under the hypotheses of Theorem 2,*

$$\Delta t \sum_{n=1}^{N-1} \|D_t^{\mathrm{BDF2}} e_h^{n+1}\|_{-1,h}^2 \le C\, h^{-2}(h^{2k} + \Delta t^4),$$

$$\textit{hence} \qquad \Big(\Delta t \sum_{n=1}^{N-1} \|p^{n+1} - p_h^{n+1}\|_0^2\Big)^{1/2} \le C\Big(h^{k-1} + \frac{\Delta t^2}{h}\Big).$$

Proof. Since $e_h = \eta - e_u$, $g := D_t^{\mathrm{BDF2}} e_h^{n+1} = \rho^{n+1} - D_t^{\mathrm{BDF2}} e_u^{n+1} \in V_h^{\mathrm{div}}$. (i) For $g \in V_h^{\mathrm{div}}$ we may take $v_h = g$ in the divergence-free dual norm:

$$\|g\|_0^2 \le \|g\|_{-1,h,\mathrm{div}} \|\nabla g\|_0 \le C_{\mathrm{inv}} h^{-1} \|g\|_{-1,h,\mathrm{div}} \|g\|_0, \qquad \text{so} \qquad \|g\|_0 \le C_{\mathrm{inv}} h^{-1} \|g\|_{-1,h,\mathrm{div}}.$$

(ii) Trivially $\|g\|_{-1,h} \le C_P \|g\|_0$ (Poincaré). (iii)

$$\|g\|_{-1,h,\mathrm{div}} \le \|\rho^{n+1}\|_{-1,h} + \|D_t^{\mathrm{BDF2}} e_u^{n+1}\|_{-1,h,\mathrm{div}}.$$

Squaring, summing, and using Lemma 5 and $\Delta t \sum \|\rho^{n+1}\|_0^2 \le C h^{2k+2}$ gives the first claim. For the second, $\|D_t^{\mathrm{BDF2}} e_u\|_{-1,h} \le \|\rho\|_{-1,h} + \|D_t^{\mathrm{BDF2}} e_h\|_{-1,h}$ is inserted in the cumulative bound of Theorem 3, whose remaining terms are $O(h^{2k} + \Delta t^4)$. □

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